MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

3640 LOW GRADE CONDITIONING, LEACHING & ADSROBTION

MISC FOUNDATION DESIGN CALCULATION

AS BUILT

MD-216-3640-EG-CV-CAL-1327

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01			Outotec / L&T Revision No.:	0	1	2	
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 306 :2007	Saudi Steel Structures Code
AISC LRFD	Building Code Requirement of Steel Structural Building
AISC Steel Design Guide 1 - 2nd Edition	Base Plate and Anchor Rod Design

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The document is prepared for design of MISC CIVIL Foundation

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The document is prepared for design of MISC CIVIL Foundation
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
3D- Model and analysis have been performed in Staad-Pro Software.
Foundation has been design using MS-Excel as Per SBC-303:2007, SBC-304:2018 & ACI 318-08.

4 DESIGN OF RCC ELEMENTS

4.1 Design — Design of RCC Str. and its components shall be designed in accordance with

[1] SBC-301-2007-Structural Loading & Forces.
[2] SBC-303-2007-Structural Soil & Foundations.
[3] SBC-304-2018-Structural Concrete Structures.
[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.
[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete
[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.
[7] ACI 318-08 Building Code Requirements.
[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

4.2 Materials

Compressive Strength of Concrete at 28th days (fc') = 35 MPa
Yield Strength of Steel (fy) = 420 MPa

4.3 Reference — Reference is made to codes, standards and specifications as indicated in Civil, Structural and Dwg. No. MD-216-39WL-EG-CV-GAD-1706-1710

5 LOAD CALCULATONS

For Load Calcultaion Ref Respective Foundation Load Data Sheets

5.1 Load Combinations

5.1.1 Factored Load Combination for Structural design (As Per Section 2.3 of SBC 301-2007)

1.1.4 DL
2. 1.4 DL + 1.7 LL+ 0.5 LR
3. 1.2 DL+ 1.6 WL + 1.0 LL + 0.5 LR
4. 1.2 DL +1.0 EL + 1.0 LL
5. 1.2 DL + 1.2 TL+ 1.6 LL
6. 0.9 DL + 1.6 WL
7. 0.9 DL + 1.0 EL

5.1.2 Load Combination for Servicibilty check

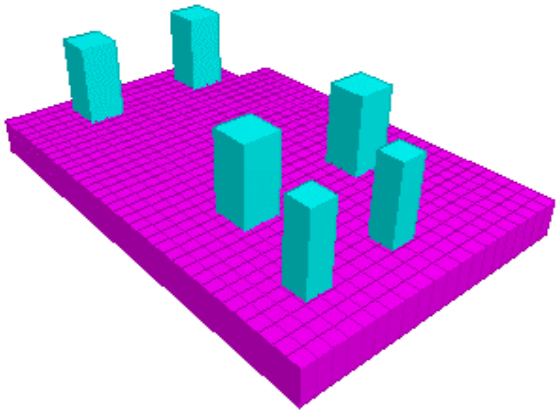
1.DL
2. DL + TL + LL
3. DL + 0.75 LL +0.75 WL
4 . DL + 0.7 E+ 0.75 L
5 .0.6 DL +1.0 WL
6. 0.6 DL + 0.7 EL

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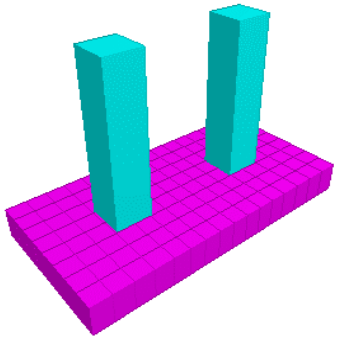
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6 GEOMETRY

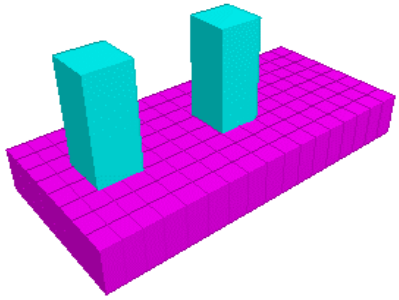
6.1 Geometry of the MISC Foundation



3D-View FDN M.K.D F4



3D-View FDN M.K.D F3



3D-View FDN M.K.D F1 & F2

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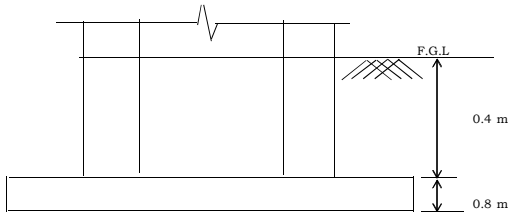
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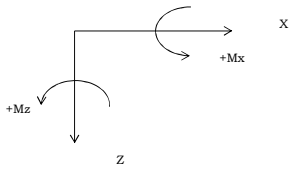
7 DESIGN OF RCC ELEMENTS

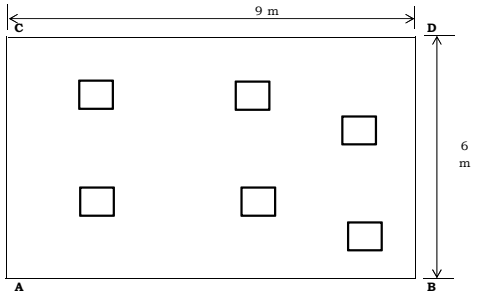
All Major components i.e. RCC Column, Beam have been design in Staad-Pro.
For Design Results, Refer Attachemnt 3 - Staad Output file.
Foundation has been design using MS-Excel as Per SBC-303:2007, SBC-304:2007 & ACI 318-08.

7.1 DESIGN OF F4 FOUNDATION :

Net Safe Bearing Capacity of Soil =	45	Mton/m^2	
Depth of Foundation from FGL =	1.2	m	
Gross Allowable Bearing Capacity of Soil =	45 + 1.2 X 1.9 =		47.28 Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	9	m
Width of Foundation (Lz) =	6	m
Thickness of Base Raft =	0.8	m
Length of Column (lx) =	0.5	m
Width of Column (lz) =	0.5	m

Area , A = Lx*Lz
= 54 m^2
Zxx = Lx*Lz^2/6
= 54 m^3
Zzz = Lz*Lx^2/6
= 81 m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	9X6X0.8X2.5 =	108
Column =	0.5X0.5X0.4X2.5 x 6=	1.5
Soil =	((9X6)-(0.5X0.5 X 6))X0.4X1.9 =	39.9
	Total =	149.4

BASE PRESSURE CHECK:

CALCULATION FOR SUBGRADE MODULAS

Settlement allowable	Δ_{allow}	=	25	mm
Factor of safety	SF	=	2.5	
Net Bearing Capacity of Soil	q_{net}	=	450	kN/m^2
Subgrade Modulas	K = SF x (q_{net}/Δ_{allow})	=	4500	T/m^2/m
Gross Bearing Capacity of Soil (With Wind Case)	$(q_{net*1.25)\ +\ soil}$	=	590	kN/m^2
Gross Bearing Capacity of Soil (Without Wind Cas	$(q_{net)\ +\ soil}$	=	477	kN/m^2

Base Pressure Summary obtained from STAAD Results

	Node	L/C	Fy MTon/m2		
Max Px	271	11 GENERATED SERVICIBILITY 1	3.119		
Min Px	271	11 GENERATED SERVICIBILITY 1	3.119		
Max Py	354	15 GENERATED SERVICIBILITY 5	13.422	<	47.7 MTon/m^2
Min Py	1090	21 GENERATED SERVICIBILITY 11	-8.195		**Safe**
Max Pz	271	11 GENERATED SERVICIBILITY 1	3.119		
Min Pz	271	11 GENERATED SERVICIBILITY 1	3.119		

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STAAD output:- Raft Thickness = 800 mm
Characteristic strength of concrete, f'_c = 35 MPa
Yield stress of steel reinforcement, f_y = 420 MPa
Clear cover for main reinforcement bars = 50mm

Raft Design

STAAD output:- Raft Thickness = 800 mm

Max BM (+) (M_y)= 478 kN m/m (MOMENT IN BOTTOM FACE)
Max BM (-) (M_y)= 417 kN m/m (MOMENT IN TOP FACE)
Max BM (+) (M_x)= 347 kN m/m (MOMENT IN BOTTOM FACE)
Max BM (-) (M_x)= 254 kN m/m (MOMENT IN TOP FACE)

Max Shear Qx = 0.362859 N/mm^2
Max Shear Qy = 0.284403 N/mm^2

Effective depth of Foundation in X direction = 740 mm
Effective depth of Foundation in Z direction = 722 mm

Moment	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
My(+)	478.00	0.87	1.16	0.0028	2080.89	1332.00	D20	150.97	300	150	2094.40
My(-)	417.00	0.76	1.02	0.0024	1810.57	1332.00	D20	173.51	300	150	2094.40
Mx(+)	347.00	0.67	0.89	0.0021	1531.92	1299.60	D20	205.08	300	200	1570.80
Mx(-)	254.00	0.49	0.65	0.0015	1116.75	1299.60	D16	154.71	300	150	1340.41

Check against one-way shear force

Allowable shear stress, (ϕ Vc)/bd = 0.74 N/mm^2
Shear stress at a distance 'd' from raft top , τ_v= 0.36 N/mm^2 **OK**

SUMMARY OF REINFORCEMENT IN RAFT: **800 mm**

(My(+) REIN. IN TOP FACE) **D20 150 c/c**
(My(-) REIN. IN BOTTOM FACE) **D20 150 c/c**
(MX(+) REIN. IN TOP FACE) **D20 200 c/c**
(MX(-)REIN. IN BOTTOM FACE) **D16 150 c/c**

CHECK AGAINST SLIDING

Factor of safety against Sliding, $S_r = \frac{(P + W_c) * \mu}{H}$

where : P = Vertical Load (kN)
W_c = Foundation weight (kN)
H = Horizontal force (kN)
μ = Concrete to soil coefficient of friction = 0.5

$P+W_c$	$(P+W_c)*\mu$	H	S_r	Allowable ($S_{r)}$	Remarks
(T)	(T)	(T)			
184.00	92.00	30.00	3.07	1.50	Pass

CHECK AGAINST OVERTURNING MOMENT

M_R = Resisting moment
M_O = Overturning moment
F_s = Factor of safety against overturning = M_R / M_O
M_R = (P + W_c + Soil) * Lever arm
M_O = (M_{ZZ} + H_X*d_t) or (M_{XX} + H_Z*d_t)

Load Comb	$P+W_c$+ Soil	Lever arm	M_R	M_O	F_S	Allowable F_S	Remarks
	(T)	(m)	(Tm)	(Tm)			
1	184.0	3	552	365.00	1.51	1.5	Pass

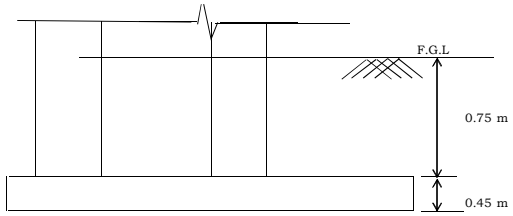
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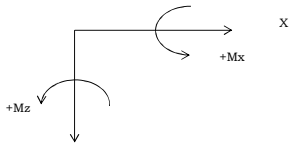
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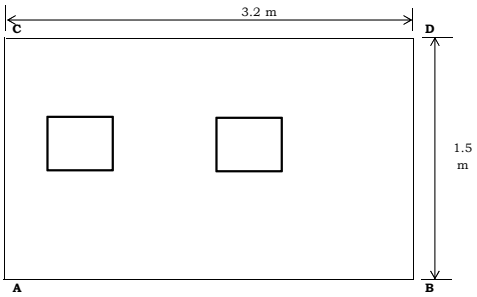
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7.1 DESIGN OF F1-F2 FOUNDATION :

Net Safe Bearing Capacity of Soil =	45	Mton/m^2	
Depth of Foundation from FGL =	1.2	m	
Gross Allowable Bearing Capacity of Soil =	45 + 1.2 X 1.9 =		47.28 Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	3.2	m
Width of Foundation (Lz) =	1.5	m
Thickness of Base Raft =	0.45	m
Length of Column (lx) =	0.4	m
Width of Column (lz) =	0.4	m

Area , A = Lx*Lz
= 4.8 m^2
Zxx = Lx*Lz^2/6
= 1.2 m^3
Zzz = Lz*Lx^2/6
= 2.56 m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	3.2X1.5X0.45X2.5 =	5.4
Column =	0.4X0.4X0.75X2.5 x 2 =	0.6
Soil =	((3.2X1.5)-(0.4X0.4X2))X0.75X1.9 =	6.384
	Total =	12.384

BASE PRESSURE CHECK:

CALCULATION FOR SUBGRADE MODULAS

Settlement allowable	Δ_{allow}	=	25	mm
Factor of safety	SF	=	2.5	
Net Bearing Capacity of Soil	q_{net}	=	450	kN/m^2
Subgrade Modulas	K = SF x (q_{net}/Δ_{allow})	=	4500	T/m^2/m
Gross Bearing Capacity of Soil (With Wind Case)	$(q_{net*1.25}) + soil$	=	590	kN/m^2
Gross Bearing Capacity of Soil (Without Wind Cas	$(q_{net}) + soil$	=	477	kN/m^2

Base Pressure Summary obtained from STAAD Results

	Node	L/C	Fy MTon/m2
Max Px	271	11 GENERATED SERVICIBILITY 1	5.706
Min Px	271	11 GENERATED SERVICIBILITY 1	5.706
Max Py	276	15 GENERATED SERVICIBILITY 5	20.327
Min Py	282	15 GENERATED SERVICIBILITY 5	-7.587
Max Pz	271	11 GENERATED SERVICIBILITY 1	5.706
Min Pz	271	11 GENERATED SERVICIBILITY 1	5.706

< 47.7 MTon/m^2
Safe

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	10/20/2020	20/11/2020	23.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	TG	TG	TG
Checked	AVG	AVG	AVG
Approved	SJM	SJM	SJM

Outotec / L&T Revision No.:	0	1	2
Date	10/20/2020	20/11/2020	23.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	TG	TG	TG
Checked	AVG	AVG	AVG
Approved	SJM	SJM	SJM

STAAD output:- Raft Thickness = 450 mm

Characteristic strength of concrete, f'_c = 35 MPa

Yield stress of steel reinforcement, f_y = 420 MPa

Clear cover for main reinforcement bars = 50mm

Raft Design

STAAD output:- Raft Thickness = 450 mm

Max BM (+) (M_y) = 57 kN m/m (MOMENT IN BOTTOM FACE)

Max BM (-) (M_y) = 39 kN m/m (MOMENT IN TOP FACE)

Max BM (+) (M_x) = 110 kN m/m (MOMENT IN BOTTOM FACE)

Max BM (-) (M_x) = 91 kN m/m (MOMENT IN TOP FACE)

Max Shear Qx = 0.215754 N/mm^2

Max Shear Qy = 0.529578 N/mm^2

Effective depth of Foundation in X direction = 394 mm

Effective depth of Foundation in Z direction = 380 mm

Moment	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
My(+)	57.00	0.37	0.49	0.0012	467.89	709.20	D12	159.47	300	150	753.98
My(-)	39.00	0.25	0.33	0.0008	319.28	709.20	D12	159.47	300	150	753.98
Mx(+)	110.00	0.76	1.02	0.0025	955.58	684.00	D16	210.41	300	150	1340.41
Mx(-)	91.00	0.63	0.84	0.0021	788.03	684.00	D16	255.14	300	200	1005.31

Check against one-way shear force

Allowable shear stress, (ϕ Vc)/bd = 0.74 N/mm^2

Shear stress at a distance 'd' from raft top , τ_v= 0.53 N/mm^2 **OK**

SUMMARY OF REINFORCEMENT IN RAFT: **450 mm**

(My(+) REIN. IN TOP FACE) **D12 150 c/c**

(My(-) REIN. IN BOTTOM FACE) **D12 150 c/c**

(MX(+) REIN. IN TOP FACE) **D16 150 c/c**

(MX(-)REIN. IN BOTTOM FACE) **D16 200 c/c**

CHECK AGAINST SLIDING

Factor of safety against Sliding, $S_r = \frac{(P + W_c) * \mu}{H}$

where : P = Vertical Load (kN)

W_c = Foundation weight (kN)

H = Horizontal force (kN)

μ = Concrete to soil coefficient of friction = 0.5

$P+W_c$	$(P+W_c)*\mu$	H	S_r	Allowable ($S_{r)}$	Remarks
(T)	(T)	(T)			
20.38	10.19	5.00	2.04	1.50	Pass

CHECK AGAINST OVERTURNING MOMENT

M_R = Resisting moment

M_O = Overturning moment

F_s = Factor of safety against overturning = M_R / M_O

M_R = (P + W_c + Soil) * Lever arm

M_O = (M_{ZZ} + H_X*d_l) or (M_{XX} + H_Z*d_l)

Load Comb	$P+W_c$+ Soil	Lever arm	M_R	M_O	F_S	Allowable F_S	Remarks
	(T)	(m)	(Tm)	(Tm)			
1	20.4	1.2	24.4608	16.00	1.53	1.5	Pass

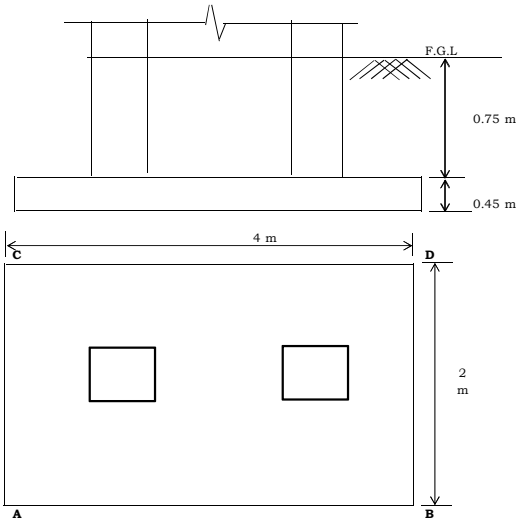
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

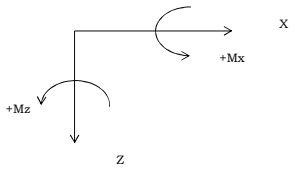
Ma'aden Revision No:	A01	C01	D01
Date	10/22/2020	20/11/2020	23.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	TG	TG	TG
Checked	AVG	AVG	AVG
Approved	SJM	SJM	SJM

Outotec / L&T Revision No.:	0	1	2
Date	10/22/2020	20/11/2020	23.11.2023
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7.1 DESIGN OF F3 FOUNDATION :

Net Safe Bearing Capacity of Soil =	45	Mton/m^2	
Depth of Foundation from FGL =	1.2	m	
Gross Allowable Bearing Capacity of Soil =	45 + 1.2 X 1.9 =		47.28 Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	





FOUNDATION DIMENSION :

Length of Foundation (Lx) =	4	m
Width of Foundation (Lz) =	2	m
Thickness of Base Raft =	0.45	m
Length of Column (lx) =	0.5	m
Width of Column (lz) =	0.5	m

Area , A = Lx*Lz
= 8 m^2
Zxx = Lx*Lz^2/6
= 2.667 m^3
Zzz = Lz*Lx^2/6
= 5.333 m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	4X2X0.45X2.5 =	9
Column =	0.5X0.5X0.75X2.5 x 2 =	0.938
Soil =	((4X2)-(0.5X0.5 X2))X0.75X1.9 =	10.688
	Total =	20.626

BASE PRESSURE CHECK:

CALCULATION FOR SUBGRADE MODULAS

Settlement allowable	Δ_{allow}	=	25	mm
Factor of safety	SF	=	2.5	
Net Bearing Capacity of Soil	q_{net}	=	450	kN/m^2
Subgrade Modulas	K = SF x (q_{net}/Δ_{allow})	=	4500	T/m^2/m
Gross Bearing Capacity of Soil (With Wind Case)	$(q_{net*1.25)\ +\ soil}$	=	590	kN/m^2
Gross Bearing Capacity of Soil (Without Wind Cas	$(q_{net)\ +\ soil}$	=	477	kN/m^2

Base Pressure Summary obtained from STAAD Results

	Node	L/C	Fy MTon/m2		
Max Px	271	11 GENERATED SERVICIBILITY 1	3.301		
Min Px	271	11 GENERATED SERVICIBILITY 1	3.301		
Max Py	280	13 GENERATED SERVICIBILITY 3	12.255	<	47.7 MTon/m^2
Min Py	279	21 GENERATED SERVICIBILITY 11	-2.21		**Safe**
Max Pz	271	11 GENERATED SERVICIBILITY 1	3.301		
Min Pz	271	11 GENERATED SERVICIBILITY 1	3.301		

Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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Ma'aden Revision No:	A01	C01	D01
Date	10/22/2020	20/11/2020	23.11.2023
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Approved	SJM	SJM	SJM

STAAD output:- Raft Thickness = 450 mm

Characteristic strength of concrete, f'_c = 35 MPa

Yield stress of steel reinforcement, f_y = 420 MPa

Clear cover for main reinforcement bars = 50mm

Raft Design

STAAD output:- Raft Thickness = 450 mm

Max BM (+) (M_y)= 46 kN m/m (MOMENT IN BOTTOM FACE)

Max BM (-) (M_y)= 20 kN m/m (MOMENT IN TOP FACE)

Max BM (+) (M_x)= 92 kN m/m (MOMENT IN BOTTOM FACE)

Max BM (-) (M_x)= 67 kN m/m (MOMENT IN TOP FACE)

Max Shear Qx = 0.147105 N/mm^2

Max Shear Qy = 0.353052 N/mm^2

Effective depth of Foundation in X direction = 394 mm

Effective depth of Foundation in Z direction = 382 mm

Moment	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
My(+)	46.00	0.30	0.40	0.0010	376.98	709.20	D12	159.47	300	150	753.98
My(-)	20.00	0.13	0.17	0.0004	163.27	709.20	D12	159.47	300	150	753.98
Mx(+)	92.00	0.63	0.84	0.0021	801.02	687.60	D16	251.01	300	200	1005.31
Mx(-)	67.00	0.46	0.61	0.0015	580.95	687.60	D12	164.48	300	150	753.98

Check against one-way shear force

Allowable shear stress, (ϕ Vc)/bd = 0.74 N/mm^2

Shear stress at a distance 'd' from raft top , τ_v= 0.35 N/mm^2 **OK**

SUMMARY OF REINFORCEMENT IN RAFT: **450 mm**

(My(+) REIN. IN TOP FACE) **D12 150 c/c**

(My(-) REIN. IN BOTTOM FACE) **D12 150 c/c**

(MX(+) REIN. IN TOP FACE) **D16 200 c/c**

(MX(-)REIN. IN BOTTOM FACE) **D12 150 c/c**

CHECK AGAINST SLIDING

Factor of safety against Sliding, $S_r = \frac{(P + W_c) * \mu}{H}$

where : P = Vertical Load (kN)

W_c = Foundation weight (kN)

H = Horizontal force (kN)

μ = Concrete to soil coefficient of friction = 0.5

$P+W_c$	$(P+W_c)*\mu$	H	S_r	Allowable ($S_{r)}$	Remarks
(T)	(T)	(T)			
24.13	12.06	4.00	3.02	1.50	Pass

CHECK AGAINST OVERTURNING MOMENT

M_R = Resisting moment

M_O = Overturning moment

F_s = Factor of safety against overturning = M_R / M_O

M_R = (P + W_c + Soil) * Lever arm

M_O = (M_{ZZ} + H_X*d_t) or (M_{XX} + H_Z*d_t)

Load Comb	$P+W_c$+ Soil	Lever arm	M_R	M_O	F_S	Allowable F_S	Remarks
	(T)	(m)	(Tm)	(Tm)			
1	24.1	2	48.252	16.00	3.02	1.5	Pass

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	22/10/2020	20/11/2020	23/11/2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	TG	TG	TG
Checked	AVG	AVG	AVG
Approved	SJM	SJM	SJM

Outotec / L&T Revision No.:	0	1	2
Date	22/10/2020	20/11/2020	23/11/2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	TG	TG	TG
Checked	AVG	AVG	AVG
Approved	SJM	SJM	SJM

ATTACHMENT 1 STAAD INPUT FILE OF MISC CIVIL FDN.

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 13-Oct-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER MTON
JOINT COORDINATES
271 20 -1.5 1.573; 272 20 0.67 1.573; 273 21.85 -1.5 1.573;
274 21.85 0.67 1.573; 277 20 -1.5 0.573; 282 18.845 -1.5 1.573;
284 18.845 -1.5 0.573; 326 23.155 -1.5 -0.432; 327 23.155 0.67 -0.432;
328 20.545 -1.5 -0.432; 329 20.545 0.67 -0.432; 330 23.155 -1.5 -5.432;
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334 18.845 -1.5 -0.432; 335 20 -1.5 -0.432; 336 21.85 -1.5 -0.432;
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349 24.845 -1.5 -0.432; 350 18.845 -1.5 -5.432; 351 20.545 -1.5 -6.307;
352 23.155 -1.5 -6.307; 354 18.845 -1.5 -6.307; 355 20 -1.5 -5.432;
356 20 -1.5 -6.307; 357 21.85 -1.5 -5.432; 358 21.85 -1.5 -6.307;
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374 21.1975 -1.5 2.693; 375 21.5238 -1.5 1.573; 376 21.5238 -1.5 2.693;
377 22.1763 -1.5 1.573; 378 22.1763 -1.5 2.693; 379 22.5025 -1.5 1.573;
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POISSON 0.17
DENSITY 2.40262
ALPHA 1e-005
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MEMBER PROPERTY
320 TO 323 PRIS YD 0.75 ZD 0.75
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
643 645 647 TO 656 658 660 662 663 665 667 669 TO 678 680 682 684 TO 697 699 -
701 703 704 706 708 710 TO 719 721 723 725 TO 738 740 742 744 745 747 749 -
751 TO 760 762 764 766 767 769 771 773 TO 782 784 786 788 TO 801 803 805 -
807 TO 868 870 872 874 TO 895 897 899 901 TO 922 924 926 928 929 931 933 -
934 TO 940 942 944 945 947 949 TO 959 961 963 TO 970 972 974 TO 990 1003 1005 -
1007 1009 1011 1013 1015 1017 1019 1021 1023 1025 1027 1029 1031 1033 1035 -
1037 1039 TO 1080 1082 1084 1086 1088 1090 1092 1094 1096 1098 1100 1102 -
1104 1106 1108 1110 1112 1114 1116 1118 1119 1121 1123 1125 1127 1129 1131 -
1133 1135 1137 1139 1141 1143 1145 1147 1149 1151 1153 1155 1157 TO 1198 -
1200 1202 1204 1206 1208 1210 1212 1214 1216 1218 1220 1222 1224 1226 1228 -
1230 1232 1234 1236 TO 1297 1299 1301 1303 1305 1307 1309 1311 1313 1315 -
1317 1319 1321 1323 1325 1327 1329 1331 1333 1335 TO 1436 1439 TO 1456 1459 -
1460 TO 1476 1482 1484 1486 1488 1490 1492 1494 1496 1498 1500 1502 1504 1506 -
1508 1510 1512 1514 1515 PLATE MAT DIRECT Y SUBGRADE 4500 PRINT
***************
LOAD 1 LOADTYPE Seismic  TITLE SLST(+X)
JOINT LOAD
```

```
272 FX 0.13
274 FX 0.05
272 FZ 0
274 FZ 0.01
272 FY -1.5
274 FY 2.51
************
329 FX 0.02
327 FX 0.26
329 FZ 0.3
327 FZ 0.19
329 FY -9.78
327 FY 5.99
************
333 FX -0.01
331 FX 0.24
333 FZ 0.04
331 FZ 0.02
333 FY -10.8
331 FY 5.82
LOAD 2 LOADTYPE Seismic  TITLE SLST(-X)
JOINT LOAD
272 FX -0.02
274 FX -0.15
272 FZ 0
274 FZ -0.15
272 FY -2.52
274 FY 12.42
************
329 FX -0.09
327 FX -0.19
329 FZ -0.35
327 FZ -0.19
329 FY -31.47
327 FY 18.02
**************
333 FX -0.06
331 FX -0.18
333 FZ 0.01
331 FZ -0.18
333 FY -30.46
331 FY 15.13
LOAD 3 LOADTYPE Dead  TITLE DEAD LOAD
SELFWEIGHT Y -1
JOINT LOAD
272 FX 0
274 FX 0
272 FZ 0
274 FZ 0
272 FY -1.33
274 FY -2.17
**************
329 FX -0.01
327 FX 0.01
329 FZ -0.01
327 FZ -0.01
329 FY -9.06
327 FY -5.84
***************
333 FX -0.02
331 FX 0.02
333 FZ 0.01
331 FZ 0.01
333 FY -10.12
331 FY -5.97
ELEMENT LOAD
643 645 647 TO 656 658 660 662 663 665 667 669 TO 678 680 682 684 TO 697 699 -
701 703 704 706 708 710 TO 719 721 723 725 TO 738 740 742 744 745 747 749 -
751 TO 760 762 764 766 767 769 771 773 TO 782 784 786 788 TO 801 803 805 -
807 TO 868 870 872 874 TO 895 897 899 901 TO 922 924 926 928 929 931 933 -
934 TO 940 942 944 945 947 949 TO 959 961 963 TO 970 972 974 TO 990 1003 1005 -
1007 1009 1011 1013 1015 1017 1019 1021 1023 1025 1027 1029 1031 1033 1035 -
1037 1039 TO 1080 1082 1084 1086 1088 1090 1092 1094 1096 1098 1100 1102 -
1104 1106 1108 1110 1112 1114 1116 1118 1119 1121 1123 1125 1127 1129 1131 -
1133 1135 1137 1139 1141 1143 1145 1147 1149 1151 1153 1155 1157 TO 1198 -
1200 1202 1204 1206 1208 1210 1212 1214 1216 1218 1220 1222 1224 1226 1228 -
1230 1232 1234 1236 TO 1297 1299 1301 1303 1305 1307 1309 1311 1313 1315 -
1317 1319 1321 1323 1325 1327 1329 1331 1333 1335 TO 1436 1439 TO 1456 1459 -
1460 TO 1476 1482 1484 1486 1488 1490 1492 1494 1496 1498 1500 1502 1504 1506 -
1508 1510 1512 1514 1515 PR GY -0.36
LOAD 4 LOADTYPE Live  TITLE LIVE LOAD
JOINT LOAD
272 FX 0.22
274 FX -0.22
```

```
272 FZ 0
274 FZ 0
272 FY -1.66
274 FY -19.47
**************
329 FX -0.07
327 FX 0.07
329 FZ -0.04
327 FZ -0.02
329 FY -39.01
327 FY -20
**************
333 FX -0.04
331 FX 0.04
333 FZ 0.04
331 FZ 0.02
333 FY -33.92
331 FY -13.25
LOAD 5 LOADTYPE Wind  TITLE WIND (+) X
JOINT LOAD
272 FX -0.75
274 FX 3.75
272 FZ 0
274 FZ -0.01
272 FY 6.57
274 FY -6.57
***************
329 FX -2.03
327 FX 14.35
329 FZ 0.06
327 FZ -0.05
329 FY 58.85
327 FY -58.85
***************
333 FX -0.42
331 FX 13.04
333 FZ -0.06
331 FZ 0.05
333 FY 61.47
331 FY -61.47
LOAD 6 LOADTYPE Wind  TITLE WIND (-) X
JOINT LOAD
272 FX 0.75
274 FX -3.75
272 FZ 0
274 FZ 0.01
272 FY -6.57
274 FY 6.57
***************
329 FX 2.01
327 FX -14.01
329 FZ -0.05
327 FZ 0.04
329 FY -57.53
327 FY 57.53
**************
333 FX 0.4
331 FX -12.71
333 FZ 0.05
331 FZ -0.04
333 FY -60
331 FY 60
LOAD 7 LOADTYPE Wind  TITLE WIND (+) Z
JOINT LOAD
272 FX 0
274 FX -0.05
272 FZ 0.02
274 FZ 0.04
272 FY -0.8
274 FY -0.45
***************
329 FX 0.23
327 FX -1
329 FZ 17.23
327 FZ 15.71
329 FY -40.06
327 FY -44.98
***************
333 FX -0.17
331 FX 0.94
333 FZ 0.27
331 FZ 0.35
333 FY 40.06
331 FY 44.98
```

```
LOAD 8 LOADTYPE Wind  TITLE WIND (-) Z
JOINT LOAD
272 FX 0
274 FX 0.04
272 FZ -0.018
274 FZ -0.039
272 FY 0.822
274 FY 0.434
**************
329 FX -0.235
327 FX 1.003
329 FZ -17.235
327 FZ -15.71
329 FY 40.063
327 FY 44.983
**************
333 FX 0.17
331 FX -0.939
333 FZ -0.272
331 FZ -0.348
333 FY -40.063
331 FY -44.983
LOAD 9 LOADTYPE Temperature  TITLE THERMAL LOAD
JOINT LOAD
272 FX 0
274 FX -3.39
272 FZ -0.14
274 FZ -0.56
272 FY -2.62
274 FY 2.62
***************
329 FX -0.17
327 FX 0.16
329 FZ -0.15
327 FZ 0.35
329 FY -0.01
327 FY 0.01
**************
333 FX -0.19
331 FX 0.19
333 FZ 0.15
331 FZ -0.35
333 FY 0.01
331 FY -0.01
LOAD 10 LOADTYPE Temperature  TITLE THERMAL LOAD
JOINT LOAD
272 FX 0
274 FX 3.39
272 FZ 0.14
274 FZ 0.56
272 FY 2.62
274 FY -2.62
**************
329 FX 0.17
327 FX -0.16
329 FZ 0.15
327 FZ -0.35
329 FY 0.01
327 FY -0.01
**************
333 FX 0.19
331 FX -0.19
333 FZ -0.15
331 FZ 0.35
333 FY -0.01
331 FY 0.01
LOAD COMB 11 GENERATED SERVICIBILITY  1
3 1.0
LOAD COMB 12 GENERATED SERVICIBILITY  2
3 1.0 4 1.0 9 1.0
LOAD COMB 13 GENERATED SERVICIBILITY  3
3 1.0 4 1.0 10 1.0
LOAD COMB 14 GENERATED SERVICIBILITY  4
3 1.0 4 0.75 5 0.75
LOAD COMB 15 GENERATED SERVICIBILITY  5
3 1.0 4 0.75 6 0.75
LOAD COMB 16 GENERATED SERVICIBILITY  6
3 1.0 4 0.75 7 0.75
LOAD COMB 17 GENERATED SERVICIBILITY  7
3 1.0 4 0.75 8 0.75
LOAD COMB 18 GENERATED SERVICIBILITY  8
3 1.0 4 0.75 1 0.7
LOAD COMB 19 GENERATED SERVICIBILITY  9
3 1.0 4 0.75 2 0.7
```

```
LOAD COMB 20 GENERATED SERVICIBILITY  10
3 0.6 5 1.0
LOAD COMB 21 GENERATED SERVICIBILITY  11
3 0.6 6 1.0
LOAD COMB 22 GENERATED SERVICIBILITY  12
3 0.6 7 1.0
LOAD COMB 23 GENERATED SERVICIBILITY  13
3 0.6 8 1.0
LOAD COMB 24 GENERATED SERVICIBILITY  14
3 0.6 1 0.7
LOAD COMB 25 GENERATED SERVICIBILITY  15
3 0.6 2 0.7
LOAD COMB 26 GENERATED LOAD_STRENGHT  1
3 1.4
LOAD COMB 27 GENERATED LOAD_STRENGHT  2
3 1.4 4 1.7
LOAD COMB 28 GENERATED LOAD_STRENGHT  3
3 1.2 4 1.0 5 1.6
LOAD COMB 29 GENERATED LOAD_STRENGHT  4
3 1.2 4 1.0 6 1.6
LOAD COMB 30 GENERATED LOAD_STRENGHT  5
3 1.2 4 1.0 7 1.6
LOAD COMB 31 GENERATED LOAD_STRENGHT  6
3 1.2 4 1.0 8 1.6
LOAD COMB 32 GENERATED LOAD_STRENGHT  7
3 1.2 4 1.0 1 1.0
LOAD COMB 33 GENERATED LOAD_STRENGHT  8
3 1.2 4 1.0 2 1.0
LOAD COMB 34 GENERATED LOAD_STRENGHT  9
3 0.9 5 1.6
LOAD COMB 35 GENERATED LOAD_STRENGHT  10
3 0.9 6 1.6
LOAD COMB 36 GENERATED LOAD_STRENGHT  11
3 0.9 7 1.6
LOAD COMB 37 GENERATED LOAD_STRENGHT  12
3 0.9 8 1.6
LOAD COMB 38 GENERATED LOAD_STRENGHT  13
3 0.9 1 1.0
LOAD COMB 39 GENERATED LOAD_STRENGHT  14
3 0.9 2 1.0
LOAD COMB 40 GENERATED LOAD_STRENGHT  15
3 1.2 4 1.6 9 1.2
LOAD COMB 41 GENERATED LOAD_STRENGHT  16
3 1.2 4 1.6 10 1.2
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 3500 MEMB 279 280 320 TO 323
CLB 0.04 MEMB 279 280 320 TO 323
CLS 0.04 ALL
CLT 0.04 MEMB 279 280 320 TO 323
MINSEC 10 ALL
MINMAIN 16 ALL
DESIGN COLUMN 279 280 320 TO 323
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 13-Oct-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER MTON
JOINT COORDINATES
271 20 -1.5 1.573; 272 20 -0.23 1.573; 273 21.2 -1.5 1.573;
274 21.2 -0.23 1.573; 275 19.5 -1.5 1.573; 276 19.5 -1.5 2.323;
277 19.5 -1.5 0.823; 278 22.7 -1.5 1.573; 280 22.7 -1.5 2.323;
282 22.7 -1.5 0.823; 284 20 -1.5 2.323; 286 21.2 -1.5 2.323; 288 20 -1.5 0.823;
290 21.2 -1.5 0.823; 291 20 -1.5 1.0105; 292 19.5 -1.5 1.0105;
293 20 -1.5 1.198; 294 19.5 -1.5 1.198; 295 20 -1.5 1.3855;
296 19.5 -1.5 1.3855; 297 19.6667 -1.5 0.823; 298 19.6667 -1.5 1.0105;
299 19.8333 -1.5 0.823; 300 19.8333 -1.5 1.0105; 301 19.6667 -1.5 1.198;
302 19.8333 -1.5 1.198; 303 19.6667 -1.5 1.3855; 304 19.8333 -1.5 1.3855;
305 19.6667 -1.5 1.573; 306 19.8333 -1.5 1.573; 307 19.6667 -1.5 1.7605;
308 19.5 -1.5 1.7605; 309 19.8333 -1.5 1.7605; 310 20 -1.5 1.7605;
311 19.6667 -1.5 1.948; 312 19.5 -1.5 1.948; 313 19.8333 -1.5 1.948;
314 20 -1.5 1.948; 315 19.6667 -1.5 2.1355; 316 19.5 -1.5 2.1355;
317 19.8333 -1.5 2.1355; 318 20 -1.5 2.1355; 319 19.6667 -1.5 2.323;
320 19.8333 -1.5 2.323; 321 20.2 -1.5 0.823; 322 20.2 -1.5 1.0105;
323 20.4 -1.5 0.823; 324 20.4 -1.5 1.0105; 325 20.6 -1.5 0.823;
326 20.6 -1.5 1.0105; 327 20.8 -1.5 0.823; 328 20.8 -1.5 1.0105;
329 21 -1.5 0.823; 330 21 -1.5 1.0105; 331 21.2 -1.5 1.0105;
332 20.2 -1.5 1.198; 333 20.4 -1.5 1.198; 334 20.6 -1.5 1.198;
335 20.8 -1.5 1.198; 336 21 -1.5 1.198; 337 21.2 -1.5 1.198;
338 20.2 -1.5 1.3855; 339 20.4 -1.5 1.3855; 340 20.6 -1.5 1.3855;
341 20.8 -1.5 1.3855; 342 21 -1.5 1.3855; 343 21.2 -1.5 1.3855;
344 20.2 -1.5 1.573; 345 20.4 -1.5 1.573; 346 20.6 -1.5 1.573;
347 20.8 -1.5 1.573; 348 21 -1.5 1.573; 349 20.2 -1.5 1.7605;
350 20.4 -1.5 1.7605; 351 20.6 -1.5 1.7605; 352 20.8 -1.5 1.7605;
353 21 -1.5 1.7605; 354 21.2 -1.5 1.7605; 355 20.2 -1.5 1.948;
356 20.4 -1.5 1.948; 357 20.6 -1.5 1.948; 358 20.8 -1.5 1.948;
359 21 -1.5 1.948; 360 21.2 -1.5 1.948; 361 20.2 -1.5 2.1355;
362 20.4 -1.5 2.1355; 363 20.6 -1.5 2.1355; 364 20.8 -1.5 2.1355;
365 21 -1.5 2.1355; 366 21.2 -1.5 2.1355; 367 20.2 -1.5 2.323;
368 20.4 -1.5 2.323; 369 20.6 -1.5 2.323; 370 20.8 -1.5 2.323;
371 21 -1.5 2.323; 372 21.3875 -1.5 0.823; 373 21.3875 -1.5 1.0105;
374 21.575 -1.5 0.823; 375 21.575 -1.5 1.0105; 376 21.7625 -1.5 0.823;
377 21.7625 -1.5 1.0105; 378 21.95 -1.5 0.823; 379 21.95 -1.5 1.0105;
380 22.1375 -1.5 0.823; 381 22.1375 -1.5 1.0105; 382 22.325 -1.5 0.823;
383 22.325 -1.5 1.0105; 384 22.5125 -1.5 0.823; 385 22.5125 -1.5 1.0105;
386 22.7 -1.5 1.0105; 387 21.3875 -1.5 1.198; 388 21.575 -1.5 1.198;
389 21.7625 -1.5 1.198; 390 21.95 -1.5 1.198; 391 22.1375 -1.5 1.198;
392 22.325 -1.5 1.198; 393 22.5125 -1.5 1.198; 394 22.7 -1.5 1.198;
395 21.3875 -1.5 1.3855; 396 21.575 -1.5 1.3855; 397 21.7625 -1.5 1.3855;
398 21.95 -1.5 1.3855; 399 22.1375 -1.5 1.3855; 400 22.325 -1.5 1.3855;
401 22.5125 -1.5 1.3855; 402 22.7 -1.5 1.3855; 403 21.3875 -1.5 1.573;
404 21.575 -1.5 1.573; 405 21.7625 -1.5 1.573; 406 21.95 -1.5 1.573;
407 22.1375 -1.5 1.573; 408 22.325 -1.5 1.573; 409 22.5125 -1.5 1.573;
410 21.3875 -1.5 1.7605; 411 21.575 -1.5 1.7605; 412 21.7625 -1.5 1.7605;
413 21.95 -1.5 1.7605; 414 22.1375 -1.5 1.7605; 415 22.325 -1.5 1.7605;
416 22.5125 -1.5 1.7605; 417 22.7 -1.5 1.7605; 418 21.3875 -1.5 1.948;
419 21.575 -1.5 1.948; 420 21.7625 -1.5 1.948; 421 21.95 -1.5 1.948;
422 22.1375 -1.5 1.948; 423 22.325 -1.5 1.948; 424 22.5125 -1.5 1.948;
425 22.7 -1.5 1.948; 426 21.3875 -1.5 2.1355; 427 21.575 -1.5 2.1355;
428 21.7625 -1.5 2.1355; 429 21.95 -1.5 2.1355; 430 22.1375 -1.5 2.1355;
431 22.325 -1.5 2.1355; 432 22.5125 -1.5 2.1355; 433 22.7 -1.5 2.1355;
434 21.3875 -1.5 2.323; 435 21.575 -1.5 2.323; 436 21.7625 -1.5 2.323;
437 21.95 -1.5 2.323; 438 22.1375 -1.5 2.323; 439 22.325 -1.5 2.323;
440 22.5125 -1.5 2.323;
MEMBER INCIDENCES
279 271 272; 280 273 274;
ELEMENT INCIDENCES SHELL
291 277 297 298 292; 292 297 299 300 298; 293 299 288 291 300;
294 292 298 301 294; 295 298 300 302 301; 296 300 291 293 302;
297 294 301 303 296; 298 301 302 304 303; 299 302 293 295 304;
300 296 303 305 275; 301 303 304 306 305; 302 304 295 271 306;
303 275 305 307 308; 304 305 306 309 307; 305 306 271 310 309;
306 308 307 311 312; 307 307 309 313 311; 308 309 310 314 313;
309 312 311 315 316; 310 311 313 317 315; 311 313 314 318 317;
312 316 315 319 276; 313 315 317 320 319; 314 317 318 284 320;
315 288 321 322 291; 316 321 323 324 322; 317 323 325 326 324;
318 325 327 328 326; 319 327 329 330 328; 320 329 290 331 330;
321 291 322 332 293; 322 322 324 333 332; 323 324 326 334 333;
324 326 328 335 334; 325 328 330 336 335; 326 330 331 337 336;
327 293 332 338 295; 328 332 333 339 338; 329 333 334 340 339;
330 334 335 341 340; 331 335 336 342 341; 332 336 337 343 342;
333 295 338 344 271; 334 338 339 345 344; 335 339 340 346 345;
336 340 341 347 346; 337 341 342 348 347; 338 342 343 273 348;
339 271 344 349 310; 340 344 345 350 349; 341 345 346 351 350;
342 346 347 352 351; 343 347 348 353 352; 344 348 273 354 353;
345 310 349 355 314; 346 349 350 356 355; 347 350 351 357 356;
```

```
348 351 352 358 357; 349 352 353 359 358; 350 353 354 360 359;
351 314 355 361 318; 352 355 356 362 361; 353 356 357 363 362;
354 357 358 364 363; 355 358 359 365 364; 356 359 360 366 365;
357 318 361 367 284; 358 361 362 368 367; 359 362 363 369 368;
360 363 364 370 369; 361 364 365 371 370; 362 365 366 286 371;
363 290 372 373 331; 364 372 374 375 373; 365 374 376 377 375;
366 376 378 379 377; 367 378 380 381 379; 368 380 382 383 381;
369 382 384 385 383; 370 384 282 386 385; 371 331 373 387 337;
372 373 375 388 387; 373 375 377 389 388; 374 377 379 390 389;
375 379 381 391 390; 376 381 383 392 391; 377 383 385 393 392;
378 385 386 394 393; 379 337 387 395 343; 380 387 388 396 395;
381 388 389 397 396; 382 389 390 398 397; 383 390 391 399 398;
384 391 392 400 399; 385 392 393 401 400; 386 393 394 402 401;
387 343 395 403 273; 388 395 396 404 403; 389 396 397 405 404;
390 397 398 406 405; 391 398 399 407 406; 392 399 400 408 407;
393 400 401 409 408; 394 401 402 278 409; 395 273 403 410 354;
396 403 404 411 410; 397 404 405 412 411; 398 405 406 413 412;
399 406 407 414 413; 400 407 408 415 414; 401 408 409 416 415;
402 409 278 417 416; 403 354 410 418 360; 404 410 411 419 418;
405 411 412 420 419; 406 412 413 421 420; 407 413 414 422 421;
408 414 415 423 422; 409 415 416 424 423; 410 416 417 425 424;
411 360 418 426 366; 412 418 419 427 426; 413 419 420 428 427;
414 420 421 429 428; 415 421 422 430 429; 416 422 423 431 430;
417 423 424 432 431; 418 424 425 433 432; 419 366 426 434 286;
420 426 427 435 434; 421 427 428 436 435; 422 428 429 437 436;
423 429 430 438 437; 424 430 431 439 438; 425 431 432 440 439;
426 432 433 280 440;
ELEMENT PROPERTY
291 TO 426 THICKNESS 0.45
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.21467e+006
POISSON 0.17
DENSITY 2.40262
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 2812.28
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
279 280 PRIS YD 0.4 ZD 0.4
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
291 TO 426 PLATE MAT DIRECT Y SUBGRADE 4500 PRINT
***************
LOAD 1 LOADTYPE Seismic  TITLE SLST(+X)
JOINT LOAD
272 FX 0.09
274 FX 0.03
272 FZ 0.06
274 FZ 0.04
272 FY 0.23
274 FY -0.22
LOAD 2 LOADTYPE Seismic  TITLE SLST(-X)
JOINT LOAD
272 FX -0.09
274 FX -0.03
272 FZ -0.06
274 FZ -0.03
272 FY -0.23
274 FY 0.22
LOAD 3 LOADTYPE Dead  TITLE DEAD LOAD
SELFWEIGHT Y -1
JOINT LOAD
272 FX 0.02
274 FX -0.02
272 FZ 0.75
274 FZ 1.68
272 FY -3.85
274 FY -5.11
ELEMENT LOAD
291 TO 426 PR GY -1.425
LOAD 4 LOADTYPE Live  TITLE LIVE LOAD
JOINT LOAD
272 FX 0.05
274 FX -0.06
272 FZ 1.3
274 FZ 4.04
272 FY -2.9
274 FY -5.66
LOAD 5 LOADTYPE Wind  TITLE WIND (+) X
JOINT LOAD
272 FX 4.6
```

```
274 FX -1.36
272 FZ 0.15
274 FZ -0.19
272 FY 10.42
274 FY -10.39
LOAD 6 LOADTYPE Wind  TITLE WIND (-) X
JOINT LOAD
272 FX -4.82
274 FX 1.45
272 FZ -0.17
274 FZ 0.21
272 FY -10.69
274 FY 10.66
LOAD 7 LOADTYPE Wind  TITLE WIND (+) Z
JOINT LOAD
272 FX -0.01
274 FX 0
272 FZ 2.62
274 FZ 2.1
272 FY -2.78
274 FY -2.78
LOAD 8 LOADTYPE Wind  TITLE WIND (-) Z
JOINT LOAD
272 FX 0.011
274 FX -0.001
272 FZ -2.622
274 FZ -2.102
272 FY 2.778
274 FY 2.782
LOAD 9 LOADTYPE Temperature  TITLE THERMAL LOAD
JOINT LOAD
272 FX -0.26
274 FX 0.07
272 FZ -1.68
274 FZ 0.88
272 FY 1.55
274 FY -1.74
LOAD 10 LOADTYPE Temperature  TITLE THERMAL LOAD
JOINT LOAD
272 FX 0.26
274 FX -0.07
272 FZ 1.68
274 FZ -0.88
272 FY -1.55
274 FY 1.74
LOAD COMB 11 GENERATED SERVICIBILITY  1
3 1.0
LOAD COMB 12 GENERATED SERVICIBILITY  2
3 1.0 4 1.0 9 1.0
LOAD COMB 13 GENERATED SERVICIBILITY  3
3 1.0 4 1.0 10 1.0
LOAD COMB 14 GENERATED SERVICIBILITY  4
3 1.0 4 0.75 5 0.75
LOAD COMB 15 GENERATED SERVICIBILITY  5
3 1.0 4 0.75 6 0.75
LOAD COMB 16 GENERATED SERVICIBILITY  6
3 1.0 4 0.75 7 0.75
LOAD COMB 17 GENERATED SERVICIBILITY  7
3 1.0 4 0.75 8 0.75
LOAD COMB 18 GENERATED SERVICIBILITY  8
3 1.0 4 0.75 1 0.7
LOAD COMB 19 GENERATED SERVICIBILITY  9
3 1.0 4 0.75 2 0.7
LOAD COMB 20 GENERATED SERVICIBILITY  10
3 0.6 5 1.0
LOAD COMB 21 GENERATED SERVICIBILITY  11
3 0.6 6 1.0
LOAD COMB 22 GENERATED SERVICIBILITY  12
3 0.6 7 1.0
LOAD COMB 23 GENERATED SERVICIBILITY  13
3 0.6 8 1.0
LOAD COMB 24 GENERATED SERVICIBILITY  14
3 0.6 1 0.7
LOAD COMB 25 GENERATED SERVICIBILITY  15
3 0.6 2 0.7
LOAD COMB 26 GENERATED LOAD_STRENGHT  1
3 1.4
LOAD COMB 27 GENERATED LOAD_STRENGHT  2
3 1.4 4 1.7
LOAD COMB 28 GENERATED LOAD_STRENGHT  3
3 1.2 4 1.0 5 1.6
LOAD COMB 29 GENERATED LOAD_STRENGHT  4
3 1.2 4 1.0 6 1.6
LOAD COMB 30 GENERATED LOAD_STRENGHT  5
```

```
3 1.2 4 1.0 7 1.6
LOAD COMB 31 GENERATED LOAD_STRENGHT  6
3 1.2 4 1.0 8 1.6
LOAD COMB 32 GENERATED LOAD_STRENGHT  7
3 1.2 4 1.0 1 1.0
LOAD COMB 33 GENERATED LOAD_STRENGHT  8
3 1.2 4 1.0 2 1.0
LOAD COMB 34 GENERATED LOAD_STRENGHT  9
3 0.9 5 1.6
LOAD COMB 35 GENERATED LOAD_STRENGHT  10
3 0.9 6 1.6
LOAD COMB 36 GENERATED LOAD_STRENGHT  11
3 0.9 7 1.6
LOAD COMB 37 GENERATED LOAD_STRENGHT  12
3 0.9 8 1.6
LOAD COMB 38 GENERATED LOAD_STRENGHT  13
3 0.9 1 1.0
LOAD COMB 39 GENERATED LOAD_STRENGHT  14
3 0.9 2 1.0
LOAD COMB 40 GENERATED LOAD_STRENGHT  15
3 1.2 4 1.6 9 1.2
LOAD COMB 41 GENERATED LOAD_STRENGHT  16
3 1.2 4 1.6 10 1.2
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 3500 MEMB 279 280
CLB 0.04 MEMB 279 280
CLS 0.04 ALL
CLT 0.04 MEMB 279 280
MINSEC 10 ALL
MINMAIN 16 ALL
DESIGN COLUMN 279 280
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 13-Oct-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER MTON
JOINT COORDINATES
271 20 -1.5 1.573; 272 20 1.07 1.573; 273 21.85 -1.5 1.573;
274 21.85 1.07 1.573; 275 20 -1.5 2.573; 276 21.85 -1.5 2.573;
277 20 -1.5 0.573; 278 21.85 -1.5 0.573; 279 22.925 -1.5 1.573;
280 22.925 -1.5 2.573; 281 22.925 -1.5 0.573; 282 18.925 -1.5 1.573;
283 18.925 -1.5 2.573; 284 18.925 -1.5 0.573; 285 19.1937 -1.5 0.573;
286 19.1937 -1.5 0.823; 287 18.925 -1.5 0.823; 288 19.4625 -1.5 0.573;
289 19.4625 -1.5 0.823; 290 19.7313 -1.5 0.573; 291 19.7313 -1.5 0.823;
292 20 -1.5 0.823; 293 19.1937 -1.5 1.073; 294 18.925 -1.5 1.073;
295 19.4625 -1.5 1.073; 296 19.7313 -1.5 1.073; 297 20 -1.5 1.073;
298 19.1937 -1.5 1.323; 299 18.925 -1.5 1.323; 300 19.4625 -1.5 1.323;
301 19.7313 -1.5 1.323; 302 20 -1.5 1.323; 303 19.1937 -1.5 1.573;
304 19.4625 -1.5 1.573; 305 19.7313 -1.5 1.573; 306 19.1937 -1.5 1.823;
307 18.925 -1.5 1.823; 308 19.4625 -1.5 1.823; 309 19.7313 -1.5 1.823;
310 20 -1.5 1.823; 311 19.1937 -1.5 2.073; 312 18.925 -1.5 2.073;
313 19.4625 -1.5 2.073; 314 19.7313 -1.5 2.073; 315 20 -1.5 2.073;
316 19.1937 -1.5 2.323; 317 18.925 -1.5 2.323; 318 19.4625 -1.5 2.323;
319 19.7313 -1.5 2.323; 320 20 -1.5 2.323; 321 19.1937 -1.5 2.573;
322 19.4625 -1.5 2.573; 323 19.7313 -1.5 2.573; 324 22.1188 -1.5 0.573;
325 22.1188 -1.5 0.823; 326 21.85 -1.5 0.823; 327 22.3875 -1.5 0.573;
328 22.3875 -1.5 0.823; 329 22.6563 -1.5 0.573; 330 22.6563 -1.5 0.823;
331 22.925 -1.5 0.823; 332 22.1188 -1.5 1.073; 333 21.85 -1.5 1.073;
334 22.3875 -1.5 1.073; 335 22.6563 -1.5 1.073; 336 22.925 -1.5 1.073;
337 22.1188 -1.5 1.323; 338 21.85 -1.5 1.323; 339 22.3875 -1.5 1.323;
340 22.6563 -1.5 1.323; 341 22.925 -1.5 1.323; 342 22.1188 -1.5 1.573;
343 22.3875 -1.5 1.573; 344 22.6563 -1.5 1.573; 345 22.1188 -1.5 1.823;
346 21.85 -1.5 1.823; 347 22.3875 -1.5 1.823; 348 22.6563 -1.5 1.823;
349 22.925 -1.5 1.823; 350 22.1188 -1.5 2.073; 351 21.85 -1.5 2.073;
352 22.3875 -1.5 2.073; 353 22.6563 -1.5 2.073; 354 22.925 -1.5 2.073;
355 22.1188 -1.5 2.323; 356 21.85 -1.5 2.323; 357 22.3875 -1.5 2.323;
358 22.6563 -1.5 2.323; 359 22.925 -1.5 2.323; 360 22.1188 -1.5 2.573;
361 22.3875 -1.5 2.573; 362 22.6563 -1.5 2.573; 363 20.2313 -1.5 0.573;
364 20.2313 -1.5 0.823; 365 20.4625 -1.5 0.573; 366 20.4625 -1.5 0.823;
367 20.6938 -1.5 0.573; 368 20.6938 -1.5 0.823; 369 20.925 -1.5 0.573;
370 20.925 -1.5 0.823; 371 21.1563 -1.5 0.573; 372 21.1563 -1.5 0.823;
373 21.3875 -1.5 0.573; 374 21.3875 -1.5 0.823; 375 21.6187 -1.5 0.573;
376 21.6187 -1.5 0.823; 377 20.2313 -1.5 1.073; 378 20.4625 -1.5 1.073;
379 20.6938 -1.5 1.073; 380 20.925 -1.5 1.073; 381 21.1563 -1.5 1.073;
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385 20.4625 -1.5 1.323; 386 20.6938 -1.5 1.323; 387 20.925 -1.5 1.323;
388 21.1563 -1.5 1.323; 389 21.3875 -1.5 1.323; 390 21.6187 -1.5 1.323;
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394 20.925 -1.5 1.573; 395 21.1563 -1.5 1.573; 396 21.3875 -1.5 1.573;
397 21.6187 -1.5 1.573; 398 20.2313 -1.5 1.823; 399 20.4625 -1.5 1.823;
400 20.6938 -1.5 1.823; 401 20.925 -1.5 1.823; 402 21.1563 -1.5 1.823;
403 21.3875 -1.5 1.823; 404 21.6187 -1.5 1.823; 405 20.2313 -1.5 2.073;
406 20.4625 -1.5 2.073; 407 20.6938 -1.5 2.073; 408 20.925 -1.5 2.073;
409 21.1563 -1.5 2.073; 410 21.3875 -1.5 2.073; 411 21.6187 -1.5 2.073;
412 20.2313 -1.5 2.323; 413 20.4625 -1.5 2.323; 414 20.6938 -1.5 2.323;
415 20.925 -1.5 2.323; 416 21.1563 -1.5 2.323; 417 21.3875 -1.5 2.323;
418 21.6187 -1.5 2.323; 419 20.2313 -1.5 2.573; 420 20.4625 -1.5 2.573;
421 20.6938 -1.5 2.573; 422 20.925 -1.5 2.573; 423 21.1563 -1.5 2.573;
424 21.3875 -1.5 2.573; 425 21.6187 -1.5 2.573;
MEMBER INCIDENCES
279 271 272; 280 273 274;
ELEMENT INCIDENCES SHELL
287 284 285 286 287; 288 285 288 289 286; 289 288 290 291 289;
290 290 277 292 291; 291 287 286 293 294; 292 286 289 295 293;
293 289 291 296 295; 294 291 292 297 296; 295 294 293 298 299;
296 293 295 300 298; 297 295 296 301 300; 298 296 297 302 301;
299 299 298 303 282; 300 298 300 304 303; 301 300 301 305 304;
302 301 302 271 305; 303 282 303 306 307; 304 303 304 308 306;
305 304 305 309 308; 306 305 271 310 309; 307 307 306 311 312;
308 306 308 313 311; 309 308 309 314 313; 310 309 310 315 314;
311 312 311 316 317; 312 311 313 318 316; 313 313 314 319 318;
314 314 315 320 319; 315 317 316 321 283; 316 316 318 322 321;
317 318 319 323 322; 318 319 320 275 323; 319 278 324 325 326;
320 324 327 328 325; 321 327 329 330 328; 322 329 281 331 330;
323 326 325 332 333; 324 325 328 334 332; 325 328 330 335 334;
326 330 331 336 335; 327 333 332 337 338; 328 332 334 339 337;
329 334 335 340 339; 330 335 336 341 340; 331 338 337 342 273;
332 337 339 343 342; 333 339 340 344 343; 334 340 341 279 344;
335 273 342 345 346; 336 342 343 347 345; 337 343 344 348 347;
338 344 279 349 348; 339 346 345 350 351; 340 345 347 352 350;
341 347 348 353 352; 342 348 349 354 353; 343 351 350 355 356;
344 350 352 357 355; 345 352 353 358 357; 346 353 354 359 358;
347 356 355 360 276; 348 355 357 361 360; 349 357 358 362 361;
350 358 359 280 362; 351 277 363 364 292; 352 363 365 366 364;
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353 365 367 368 366; 354 367 369 370 368; 355 369 371 372 370;
356 371 373 374 372; 357 373 375 376 374; 358 375 278 326 376;
359 292 364 377 297; 360 364 366 378 377; 361 366 368 379 378;
362 368 370 380 379; 363 370 372 381 380; 364 372 374 382 381;
365 374 376 383 382; 366 376 326 333 383; 367 297 377 384 302;
368 377 378 385 384; 369 378 379 386 385; 370 379 380 387 386;
371 380 381 388 387; 372 381 382 389 388; 373 382 383 390 389;
374 383 333 338 390; 375 302 384 391 271; 376 384 385 392 391;
377 385 386 393 392; 378 386 387 394 393; 379 387 388 395 394;
380 388 389 396 395; 381 389 390 397 396; 382 390 338 273 397;
383 271 391 398 310; 384 391 392 399 398; 385 392 393 400 399;
386 393 394 401 400; 387 394 395 402 401; 388 395 396 403 402;
389 396 397 404 403; 390 397 273 346 404; 391 310 398 405 315;
392 398 399 406 405; 393 399 400 407 406; 394 400 401 408 407;
395 401 402 409 408; 396 402 403 410 409; 397 403 404 411 410;
398 404 346 351 411; 399 315 405 412 320; 400 405 406 413 412;
401 406 407 414 413; 402 407 408 415 414; 403 408 409 416 415;
404 409 410 417 416; 405 410 411 418 417; 406 411 351 356 418;
407 320 412 419 275; 408 412 413 420 419; 409 413 414 421 420;
410 414 415 422 421; 411 415 416 423 422; 412 416 417 424 423;
413 417 418 425 424; 414 418 356 276 425;
ELEMENT PROPERTY
287 TO 414 THICKNESS 0.45
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.21467e+006
POISSON 0.17
DENSITY 2.40262
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 2812.28
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
279 280 PRIS YD 0.5 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
287 TO 414 PLATE MAT DIRECT Y SUBGRADE 4500 PRINT
***************
LOAD 1 LOADTYPE Seismic  TITLE SLST(+X)
JOINT LOAD
272 FX 0.13
274 FX 0.05
272 FZ 0
274 FZ 0.01
272 FY -1.5
274 FY 2.51
LOAD 2 LOADTYPE Seismic  TITLE SLST(-X)
JOINT LOAD
272 FX -0.02
274 FX -0.15
272 FZ 0
274 FZ -0.15
272 FY -2.52
274 FY 12.42
LOAD 3 LOADTYPE Dead  TITLE DEAD LOAD
SELFWEIGHT Y -1
JOINT LOAD
272 FX 0
274 FX 0
272 FZ 0
274 FZ 0
272 FY -1.33
274 FY -2.17
ELEMENT LOAD
287 TO 414 PR GY -1.425
LOAD 4 LOADTYPE Live  TITLE LIVE LOAD
JOINT LOAD
272 FX 0.22
274 FX -0.22
272 FZ 0
274 FZ 0
272 FY -1.66
274 FY -19.47
LOAD 5 LOADTYPE Wind  TITLE WIND (+) X
JOINT LOAD
272 FX -0.75
274 FX 3.75
272 FZ 0
274 FZ -0.01
272 FY 6.57
274 FY -6.57
LOAD 6 LOADTYPE Wind  TITLE WIND (-) X
```

```
JOINT LOAD
272 FX 0.75
274 FX -3.75
272 FZ 0
274 FZ 0.01
272 FY -6.57
274 FY 6.57
LOAD 7 LOADTYPE Wind  TITLE WIND (+) Z
JOINT LOAD
272 FX 0
274 FX -0.05
272 FZ 0.02
274 FZ 0.04
272 FY 0.8
274 FY 0.45
LOAD 8 LOADTYPE Wind  TITLE WIND (-) Z
JOINT LOAD
272 FX 0
274 FX 0.04
272 FZ -0.018
274 FZ -0.039
272 FY -0.822
274 FY -0.434
LOAD 9 LOADTYPE Temperature  TITLE THERMAL LOAD
JOINT LOAD
272 FX 0
274 FX -3.39
272 FZ -0.14
274 FZ -0.56
272 FY -2.62
274 FY 2.62
LOAD 10 LOADTYPE Temperature  TITLE THERMAL LOAD
JOINT LOAD
272 FX 0
274 FX 3.39
272 FZ 0.14
274 FZ 0.56
272 FY 2.62
274 FY -2.62
LOAD COMB 11 GENERATED SERVICIBILITY  1
3 1.0
LOAD COMB 12 GENERATED SERVICIBILITY  2
3 1.0 4 1.0 9 1.0
LOAD COMB 13 GENERATED SERVICIBILITY  3
3 1.0 4 1.0 10 1.0
LOAD COMB 14 GENERATED SERVICIBILITY  4
3 1.0 4 0.75 5 0.75
LOAD COMB 15 GENERATED SERVICIBILITY  5
3 1.0 4 0.75 6 0.75
LOAD COMB 16 GENERATED SERVICIBILITY  6
3 1.0 4 0.75 7 0.75
LOAD COMB 17 GENERATED SERVICIBILITY  7
3 1.0 4 0.75 8 0.75
LOAD COMB 18 GENERATED SERVICIBILITY  8
3 1.0 4 0.75 1 0.7
LOAD COMB 19 GENERATED SERVICIBILITY  9
3 1.0 4 0.75 2 0.7
LOAD COMB 20 GENERATED SERVICIBILITY  10
3 0.6 5 1.0
LOAD COMB 21 GENERATED SERVICIBILITY  11
3 0.6 6 1.0
LOAD COMB 22 GENERATED SERVICIBILITY  12
3 0.6 7 1.0
LOAD COMB 23 GENERATED SERVICIBILITY  13
3 0.6 8 1.0
LOAD COMB 24 GENERATED SERVICIBILITY  14
3 0.6 1 0.7
LOAD COMB 25 GENERATED SERVICIBILITY  15
3 0.6 2 0.7
LOAD COMB 26 GENERATED LOAD_STRENGHT  1
3 1.4
LOAD COMB 27 GENERATED LOAD_STRENGHT  2
3 1.4 4 1.7
LOAD COMB 28 GENERATED LOAD_STRENGHT  3
3 1.2 4 1.0 5 1.6
LOAD COMB 29 GENERATED LOAD_STRENGHT  4
3 1.2 4 1.0 6 1.6
LOAD COMB 30 GENERATED LOAD_STRENGHT  5
3 1.2 4 1.0 7 1.6
LOAD COMB 31 GENERATED LOAD_STRENGHT  6
3 1.2 4 1.0 8 1.6
LOAD COMB 32 GENERATED LOAD_STRENGHT  7
3 1.2 4 1.0 1 1.0
LOAD COMB 33 GENERATED LOAD_STRENGHT  8
```

```
3 1.2 4 1.0 2 1.0
LOAD COMB 34 GENERATED LOAD_STRENGHT  9
3 0.9 5 1.6
LOAD COMB 35 GENERATED LOAD_STRENGHT  10
3 0.9 6 1.6
LOAD COMB 36 GENERATED LOAD_STRENGHT  11
3 0.9 7 1.6
LOAD COMB 37 GENERATED LOAD_STRENGHT  12
3 0.9 8 1.6
LOAD COMB 38 GENERATED LOAD_STRENGHT  13
3 0.9 1 1.0
LOAD COMB 39 GENERATED LOAD_STRENGHT  14
3 0.9 2 1.0
LOAD COMB 40 GENERATED LOAD_STRENGHT  15
3 1.2 4 1.6 9 1.2
LOAD COMB 41 GENERATED LOAD_STRENGHT  16
3 1.2 4 1.6 10 1.2
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 3500 MEMB 279 280
CLB 0.04 MEMB 279 280
CLS 0.04 ALL
CLT 0.04 MEMB 279 280
MINSEC 10 ALL
MINMAIN 16 ALL
DESIGN COLUMN 279 280
END CONCRETE DESIGN
FINISH
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	22/10/2020	20/11/2020	23.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	TG	TG	TG
Checked	AVG	AVG	AVG
Approved	SJM	SJM	SJM

Outotec / L&T Revision No.:	0	1	2
Date	22/10/2020	20/11/2020	23.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	TG	TG	TG
Checked	AVG	AVG	AVG
Approved	SJM	SJM	SJM

ATTACHMENT 2 STAAD OUTPUT FILE OF MISC CIVIL FDN.

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5           *
*           Version  20.07.10.64                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    OCT 21, 2020                 *
*           Time=    17:14:20                     *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F4-FDN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 13-OCT-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 271 20 -1.5 1.573; 272 20 0.67 1.573; 273 21.85 -1.5 1.573
     9. 274 21.85 0.67 1.573; 277 20 -1.5 0.573; 282 18.845 -1.5 1.573
    10. 284 18.845 -1.5 0.573; 326 23.155 -1.5 -0.432; 327 23.155 0.67 -0.432
    11. 328 20.545 -1.5 -0.432; 329 20.545 0.67 -0.432; 330 23.155 -1.5 -5.432
    12. 331 23.155 0.67 -5.432; 332 20.545 -1.5 -5.432; 333 20.545 0.67 -5.432
    13. 334 18.845 -1.5 -0.432; 335 20 -1.5 -0.432; 336 21.85 -1.5 -0.432
    14. 337 21.85 -1.5 0.573; 338 20.545 -1.5 0.573; 342 23.155 -1.5 1.573
    15. 343 23.155 -1.5 0.573; 344 20.545 -1.5 1.573; 348 24.845 -1.5 0.573
    16. 349 24.845 -1.5 -0.432; 350 18.845 -1.5 -5.432; 351 20.545 -1.5 -6.307
    17. 352 23.155 -1.5 -6.307; 354 18.845 -1.5 -6.307; 355 20 -1.5 -5.432
    18. 356 20 -1.5 -6.307; 357 21.85 -1.5 -5.432; 358 21.85 -1.5 -6.307
    19. 359 20 -1.5 2.693; 360 20.545 -1.5 2.693; 361 21.85 -1.5 2.693
    20. 362 23.155 -1.5 2.693; 363 24.845 -1.5 2.693; 364 18.845 -1.5 2.693
    21. 365 19.1337 -1.5 1.573; 366 19.1337 -1.5 2.693; 367 19.4225 -1.5 1.573
    22. 368 19.4225 -1.5 2.693; 369 19.7113 -1.5 1.573; 370 19.7113 -1.5 2.693
    23. 371 20.8713 -1.5 1.573; 372 20.8713 -1.5 2.693; 373 21.1975 -1.5 1.573
    24. 374 21.1975 -1.5 2.693; 375 21.5238 -1.5 1.573; 376 21.5238 -1.5 2.693
    25. 377 22.1763 -1.5 1.573; 378 22.1763 -1.5 2.693; 379 22.5025 -1.5 1.573
    26. 380 22.5025 -1.5 2.693; 381 22.8288 -1.5 1.573; 382 22.8288 -1.5 2.693
    27. 383 20.8713 -1.5 0.573; 384 21.1975 -1.5 0.573; 385 21.5238 -1.5 0.573
    28. 386 22.1763 -1.5 0.573; 387 22.5025 -1.5 0.573; 388 22.8288 -1.5 0.573
    29. 389 19.1337 -1.5 -0.432; 390 19.1337 -1.5 0.573; 391 19.4225 -1.5 -0.432
    30. 392 19.4225 -1.5 0.573; 393 19.7113 -1.5 -0.432; 394 19.7113 -1.5 0.573
    31. 395 20.8713 -1.5 -0.432; 396 21.1975 -1.5 -0.432; 397 21.5238 -1.5 -0.432
    32. 398 22.1763 -1.5 -0.432; 399 22.5025 -1.5 -0.432; 400 22.8288 -1.5 -0.432
    33. 401 19.1337 -1.5 -5.432; 402 19.4225 -1.5 -5.432; 403 19.7113 -1.5 -5.432
    34. 404 20.8713 -1.5 -5.432; 405 21.1975 -1.5 -5.432; 406 21.5238 -1.5 -5.432
    35. 407 22.1763 -1.5 -5.432; 408 22.5025 -1.5 -5.432; 409 22.8288 -1.5 -5.432
    36. 410 22.1763 -1.5 -6.307; 411 22.5025 -1.5 -6.307; 412 22.8288 -1.5 -6.307
    37. 413 20.8713 -1.5 -6.307; 414 21.1975 -1.5 -6.307; 415 21.5238 -1.5 -6.307
    38. 416 19.1337 -1.5 -6.307; 417 19.4225 -1.5 -6.307; 418 19.7113 -1.5 -6.307
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39. 419 20.2725 -1.5 1.573; 420 20.2725 -1.5 2.693; 421 20.2725 -1.5 0.573
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41. 447 23.4367 -1.5 0.573; 448 23.7183 -1.5 0.573; 449 24 -1.5 0.573
42. 450 24.2817 -1.5 0.573; 451 24.5633 -1.5 0.573; 452 23.4367 -1.5 -0.432
43. 453 23.7183 -1.5 -0.432; 454 24 -1.5 -0.432; 455 24.2817 -1.5 -0.432
44. 456 24.5633 -1.5 -0.432; 457 23.4367 -1.5 -5.432; 458 23.7183 -1.5 -5.432
45. 459 24 -1.5 -5.432; 462 24.845 -1.5 1.573; 463 23.4367 -1.5 1.573
46. 464 23.7183 -1.5 1.573; 465 24 -1.5 1.573; 466 24.2817 -1.5 1.573
47. 467 24.5633 -1.5 1.573; 468 23.4367 -1.5 2.693; 469 23.7183 -1.5 2.693
48. 470 24 -1.5 2.693; 471 24.2817 -1.5 2.693; 472 24.5633 -1.5 2.693
49. 473 23.4367 -1.5 -6.307; 474 23.7183 -1.5 -6.307; 475 24 -1.5 -6.307
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51. 481 18.845 -1.5 2.133; 482 19.1337 -1.5 2.413; 483 18.845 -1.5 2.413
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53. 487 19.7113 -1.5 1.853; 488 19.7113 -1.5 2.133; 489 19.7113 -1.5 2.413
54. 490 20 -1.5 1.853; 491 20 -1.5 2.133; 492 20 -1.5 2.413
55. 493 20.8713 -1.5 1.853; 494 20.545 -1.5 1.853; 495 20.8713 -1.5 2.133
56. 496 20.545 -1.5 2.133; 497 20.8713 -1.5 2.413; 498 20.545 -1.5 2.413
57. 499 21.1975 -1.5 1.853; 500 21.1975 -1.5 2.133; 501 21.1975 -1.5 2.413
58. 502 21.5238 -1.5 1.853; 503 21.5238 -1.5 2.133; 504 21.5238 -1.5 2.413
59. 505 21.85 -1.5 1.853; 506 21.85 -1.5 2.133; 507 21.85 -1.5 2.413
60. 508 22.1763 -1.5 1.853; 509 22.1763 -1.5 2.133; 510 22.1763 -1.5 2.413
61. 511 22.5025 -1.5 1.853; 512 22.5025 -1.5 2.133; 513 22.5025 -1.5 2.413
62. 514 22.8288 -1.5 1.853; 515 22.8288 -1.5 2.133; 516 22.8288 -1.5 2.413
63. 517 23.155 -1.5 1.853; 518 23.155 -1.5 2.133; 519 23.155 -1.5 2.413
64. 520 20.8713 -1.5 0.823; 521 20.545 -1.5 0.823; 522 20.8713 -1.5 1.073
65. 523 20.545 -1.5 1.073; 524 20.8713 -1.5 1.323; 525 20.545 -1.5 1.323
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500. 703 704 706 712 716 721 727 731 735 740 742 744 745 747 749 751 TO 760 767 -
501. 774 778 782 784 786 788 TO 793 797 801 808 TO 813 817 821 828 832 833 837 -
502. 844 848 852 856 860 TO 868 870 872 874 TO 895 897 899 901 TO 922 924 926 -
503. 928 929 931 933 934 957 TO 959 961 963 TO 970 1003 1005 1007 1009 1011 1013 -
504. 1015 1017 1019 1021 1023 1025 1027 1029 1031 1033 1035 1037 1039 TO 1080 -
505. 1088 1090 1092 1094 1096 1098 1100 1102 1104 1106 1108 1110 1112 1114 1127 -
506. 1129 1131 1133 1135 1137 1139 1141 1143 1145 1147 1149 1151 1153 -
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508. 1220 1222 1224 1226 1228 1230 1232 1234 1236 TO 1257 1261 TO 1274 -
509. 1281 TO 1294 1305 1307 1309 1311 1313 1315 1317 1319 1321 1323 1325 1327 -
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513. _PLATEBM 643 645 647 TO 656 660 662 663 665 667 669 TO 678 682 684 685 687 -
514. 688 TO 697 699 701 703 704 706 708 710 TO 719 721 723 725 727 TO 729 -
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526. 1338 TO 1356 1358 TO 1375 1378 TO 1395 1397 TO 1436 1439 TO 1456 1459 TO 1476 -
527. 1482 1484 1486 1488 1490 1492 1494 1496 1498 1500 1502 1504 1506 1508 1510 -
528. 1512 1514 1515
529. _PLATEDSHEAR 747 749 751 TO 756 865 TO 868 870 872 874 TO 895 897 899 901 -
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539. END GROUP DEFINITION
540. ELEMENT PROPERTY
541. 643 645 647 TO 656 658 660 662 663 665 667 669 TO 678 680 682 684 TO 697 699 -
542. 701 703 704 706 708 710 TO 719 721 723 725 TO 738 740 742 744 745 747 749 -
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543. 751 TO 760 762 764 766 767 769 771 773 TO 782 784 786 788 TO 801 803 805 -
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553. 1460 TO 1476 1482 1484 1486 1488 1490 1492 1494 1496 1498 1500 1502 1504 1506 -
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567. MEMBER PROPERTY AMERICAN
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569. CONSTANTS
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573. 701 703 704 706 708 710 TO 719 721 723 725 TO 738 740 742 744 745 747 749 -
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997	0.070412
998	0.070412
999	0.070412
1000	0.070412
1001	0.070412
1002	0.070412
1003	0.070412
1004	0.070412
1005	0.070412
1006	0.070412
1007	0.070412
1008	0.070412
1009	0.070412
1010	0.070413
1011	0.070413
1012	0.070412
1013	0.070413
1014	0.070413
1015	0.070413
1016	0.070413
1017	0.070413
1018	0.070413
1019	0.070413
1020	0.070413
1021	0.070413
1022	0.070413
1023	0.070413
1024	0.070413
1025	0.070413
1026	0.070413
1027	0.070413
1028	0.070413
1029	0.070413
1030	0.070413
1031	0.070412
1032	0.035213
1033	0.052819
1034	0.070425
1035	0.070425
1036	0.070425
1037	0.070425
1038	0.070425
1039	0.070425
1040	0.070425
1041	0.070425
1042	0.070425
1043	0.070425
1044	0.070425

1045	0.070425
1046	0.070425
1047	0.070425
1048	0.070425
1049	0.070425
1050	0.070425
1052	0.035206
1053	0.070413
1054	0.070413
1055	0.070413
1056	0.070413
1057	0.070413
1058	0.070413
1059	0.070413
1060	0.070413
1061	0.070413
1062	0.070413
1063	0.070413
1064	0.070413
1065	0.070413
1066	0.070413
1067	0.070413
1068	0.070413
1069	0.070412
1071	0.035206
1072	0.070412
1073	0.070412
1074	0.070412
1075	0.070412
1076	0.070412
1077	0.070412
1078	0.070412
1079	0.070412
1080	0.070412
1081	0.070412
1082	0.070412
1083	0.070412
1084	0.070412
1085	0.070412
1086	0.070413
1087	0.070413
1088	0.070412
1090	0.017606
1091	0.035212
1092	0.035212
1093	0.035212
1094	0.035212
1095	0.035212
1096	0.035212
1097	0.035212
1098	0.035212
1099	0.035212
1100	0.035212
1101	0.035212
1102	0.035212
1103	0.035212

```
     1104          0.035212
     1105          0.035212
     1106          0.035212
     1107          0.035212
    TOTAL         52.838070
586. ***************
587. LOAD 1 LOADTYPE SEISMIC  TITLE SLST(+X)
588. JOINT LOAD
589. 272 FX 0.13
590. 274 FX 0.05
591. 272 FZ 0
592. 274 FZ 0.01
593. 272 FY -1.5
594. 274 FY 2.51
595. ************
596. 329 FX 0.02
597. 327 FX 0.26
598. 329 FZ 0.3
599. 327 FZ 0.19
600. 329 FY -9.78
601. 327 FY 5.99
602. ************
603. 333 FX -0.01
604. 331 FX 0.24
605. 333 FZ 0.04
606. 331 FZ 0.02
607. 333 FY -10.8
608. 331 FY 5.82
609. LOAD 2 LOADTYPE SEISMIC  TITLE SLST(-X)
610. JOINT LOAD
611. 272 FX -0.02
612. 274 FX -0.15
613. 272 FZ 0
614. 274 FZ -0.15
615. 272 FY -2.52
616. 274 FY 12.42
617. ************
618. 329 FX -0.09
619. 327 FX -0.19
620. 329 FZ -0.35
621. 327 FZ -0.19
622. 329 FY -31.47
623. 327 FY 18.02
624. *************
625. 333 FX -0.06
626. 331 FX -0.18
627. 333 FZ 0.01
628. 331 FZ -0.18
629. 333 FY -30.46
630. 331 FY 15.13
631. LOAD 3 LOADTYPE DEAD  TITLE DEAD LOAD
632. SELFWEIGHT Y -1
633. JOINT LOAD
634. 272 FX 0
635. 274 FX 0
636. 272 FZ 0
```

```
637. 274 FZ 0
638. 272 FY -1.33
639. 274 FY -2.17
640. **************
641. 329 FX -0.01
642. 327 FX 0.01
643. 329 FZ -0.01
644. 327 FZ -0.01
645. 329 FY -9.06
646. 327 FY -5.84
647. **************
648. 333 FX -0.02
649. 331 FX 0.02
650. 333 FZ 0.01
651. 331 FZ 0.01
652. 333 FY -10.12
653. 331 FY -5.97
654. ELEMENT LOAD
655. 643 645 647 TO 656 658 660 662 663 665 667 669 TO 678 680 682 684 TO 697 699 -
656. 701 703 704 706 708 710 TO 719 721 723 725 TO 738 740 742 744 745 747 749 -
657. 751 TO 760 762 764 766 767 769 771 773 TO 782 784 786 788 TO 801 803 805 -
658. 807 TO 868 870 872 874 TO 895 897 899 901 TO 922 924 926 928 929 931 933 -
659. 934 TO 940 942 944 945 947 949 TO 959 961 963 TO 970 972 974 TO 990 1003 1005 -
660. 1007 1009 1011 1013 1015 1017 1019 1021 1023 1025 1027 1029 1031 1033 1035 -
661. 1037 1039 TO 1080 1082 1084 1086 1088 1090 1092 1094 1096 1098 1100 1102 -
662. 1104 1106 1108 1110 1112 1114 1116 1118 1119 1121 1123 1125 1127 1129 1131 -
663. 1133 1135 1137 1139 1141 1143 1145 1147 1149 1151 1153 1155 1157 TO 1198 -
664. 1200 1202 1204 1206 1208 1210 1212 1214 1216 1218 1220 1222 1224 1226 1228 -
665. 1230 1232 1234 1236 TO 1297 1299 1301 1303 1305 1307 1309 1311 1313 1315 -
666. 1317 1319 1321 1323 1325 1327 1329 1331 1333 1335 TO 1436 1439 TO 1456 1459 -
667. 1460 TO 1476 1482 1484 1486 1488 1490 1492 1494 1496 1498 1500 1502 1504 1506 -
668. 1508 1510 1512 1514 1515 PR GY -0.36
669. LOAD 4 LOADTYPE LIVE  TITLE LIVE LOAD
670. JOINT LOAD
671. 272 FX 0.22
672. 274 FX -0.22
673. 272 FZ 0
674. 274 FZ 0
675. 272 FY -1.66
676. 274 FY -19.47
677. *************
678. 329 FX -0.07
679. 327 FX 0.07
680. 329 FZ -0.04
681. 327 FZ -0.02
682. 329 FY -39.01
683. 327 FY -20
684. *************
685. 333 FX -0.04
686. 331 FX 0.04
687. 333 FZ 0.04
688. 331 FZ 0.02
689. 333 FY -33.92
690. 331 FY -13.25
691. LOAD 5 LOADTYPE WIND  TITLE WIND (+) X
692. JOINT LOAD
```

```
693. 272 FX -0.75
694. 274 FX 3.75
695. 272 FZ 0
696. 274 FZ -0.01
697. 272 FY 6.57
698. 274 FY -6.57
699. ***************
700. 329 FX -2.03
701. 327 FX 14.35
702. 329 FZ 0.06
703. 327 FZ -0.05
704. 329 FY 58.85
705. 327 FY -58.85
706. **************
707. 333 FX -0.42
708. 331 FX 13.04
709. 333 FZ -0.06
710. 331 FZ 0.05
711. 333 FY 61.47
712. 331 FY -61.47
713. LOAD 6 LOADTYPE WIND  TITLE WIND (-) X
714. JOINT LOAD
715. 272 FX 0.75
716. 274 FX -3.75
717. 272 FZ 0
718. 274 FZ 0.01
719. 272 FY -6.57
720. 274 FY 6.57
721. ***************
722. 329 FX 2.01
723. 327 FX -14.01
724. 329 FZ -0.05
725. 327 FZ 0.04
726. 329 FY -57.53
727. 327 FY 57.53
728. **************
729. 333 FX 0.4
730. 331 FX -12.71
731. 333 FZ 0.05
732. 331 FZ -0.04
733. 333 FY -60
734. 331 FY 60
735. LOAD 7 LOADTYPE WIND  TITLE WIND (+) Z
736. JOINT LOAD
737. 272 FX 0
738. 274 FX -0.05
739. 272 FZ 0.02
740. 274 FZ 0.04
741. 272 FY -0.8
742. 274 FY -0.45
743. **************
744. 329 FX 0.23
745. 327 FX -1.
746. 329 FZ 17.23
747. 327 FZ 15.71
748. 329 FY -40.06
```

```
749. 327 FY -44.98
750. ***************
751. 333 FX -0.17
752. 331 FX 0.94
753. 333 FZ 0.27
754. 331 FZ 0.35
755. 333 FY 40.06
756. 331 FY 44.98
757. LOAD 8 LOADTYPE WIND  TITLE WIND (-) Z
758. JOINT LOAD
759. 272 FX 0
760. 274 FX 0.04
761. 272 FZ -0.018
762. 274 FZ -0.039
763. 272 FY 0.822
764. 274 FY 0.434
765. **************
766. 329 FX -0.235
767. 327 FX 1.003
768. 329 FZ -17.235
769. 327 FZ -15.71
770. 329 FY 40.063
771. 327 FY 44.983
772. **************
773. 333 FX 0.17
774. 331 FX -0.939
775. 333 FZ -0.272
776. 331 FZ -0.348
777. 333 FY -40.063
778. 331 FY -44.983
779. LOAD 9 LOADTYPE TEMPERATURE  TITLE THERMAL LOAD
780. JOINT LOAD
781. 272 FX 0
782. 274 FX -3.39
783. 272 FZ -0.14
784. 274 FZ -0.56
785. 272 FY -2.62
786. 274 FY 2.62
787. ***************
788. 329 FX -0.17
789. 327 FX 0.16
790. 329 FZ -0.15
791. 327 FZ 0.35
792. 329 FY -0.01
793. 327 FY 0.01
794. **************
795. 333 FX -0.19
796. 331 FX 0.19
797. 333 FZ 0.15
798. 331 FZ -0.35
799. 333 FY 0.01
800. 331 FY -0.01
801. LOAD 10 LOADTYPE TEMPERATURE  TITLE THERMAL LOAD
802. JOINT LOAD
803. 272 FX 0
804. 274 FX 3.39
```

```
805. 272 FZ 0.14
806. 274 FZ 0.56
807. 272 FY 2.62
808. 274 FY -2.62
809. *************
810. 329 FX 0.17
811. 327 FX -0.16
812. 329 FZ 0.15
813. 327 FZ -0.35
814. 329 FY 0.01
815. 327 FY -0.01
816. *************
817. 333 FX 0.19
818. 331 FX -0.19
819. 333 FZ -0.15
820. 331 FZ 0.35
821. 333 FY -0.01
822. 331 FY 0.01
823. LOAD COMB 11 GENERATED SERVICIBILITY  1
824. 3 1.0
825. LOAD COMB 12 GENERATED SERVICIBILITY  2
826. 3 1.0 4 1.0 9 1.0
827. LOAD COMB 13 GENERATED SERVICIBILITY  3
828. 3 1.0 4 1.0 10 1.0
829. LOAD COMB 14 GENERATED SERVICIBILITY  4
830. 3 1.0 4 0.75 5 0.75
831. LOAD COMB 15 GENERATED SERVICIBILITY  5
832. 3 1.0 4 0.75 6 0.75
833. LOAD COMB 16 GENERATED SERVICIBILITY  6
834. 3 1.0 4 0.75 7 0.75
835. LOAD COMB 17 GENERATED SERVICIBILITY  7
836. 3 1.0 4 0.75 8 0.75
837. LOAD COMB 18 GENERATED SERVICIBILITY  8
838. 3 1.0 4 0.75 1 0.7
839. LOAD COMB 19 GENERATED SERVICIBILITY  9
840. 3 1.0 4 0.75 2 0.7
841. LOAD COMB 20 GENERATED SERVICIBILITY  10
842. 3 0.6 5 1.0
843. LOAD COMB 21 GENERATED SERVICIBILITY  11
844. 3 0.6 6 1.0
845. LOAD COMB 22 GENERATED SERVICIBILITY  12
846. 3 0.6 7 1.0
847. LOAD COMB 23 GENERATED SERVICIBILITY  13
848. 3 0.6 8 1.0
849. LOAD COMB 24 GENERATED SERVICIBILITY  14
850. 3 0.6 1 0.7
851. LOAD COMB 25 GENERATED SERVICIBILITY  15
852. 3 0.6 2 0.7
853. LOAD COMB 26 GENERATED LOAD_STRENGHT  1
854. 3 1.4
855. LOAD COMB 27 GENERATED LOAD_STRENGHT  2
856. 3 1.4 4 1.7
857. LOAD COMB 28 GENERATED LOAD_STRENGHT  3
858. 3 1.2 4 1.0 5 1.6
859. LOAD COMB 29 GENERATED LOAD_STRENGHT  4
860. 3 1.2 4 1.0 6 1.6
```

```
861. LOAD COMB 30 GENERATED LOAD_STRENGHT  5
862. 3 1.2 4 1.0 7 1.6
863. LOAD COMB 31 GENERATED LOAD_STRENGHT  6
864. 3 1.2 4 1.0 8 1.6
865. LOAD COMB 32 GENERATED LOAD_STRENGHT  7
866. 3 1.2 4 1.0 1 1.0
867. LOAD COMB 33 GENERATED LOAD_STRENGHT  8
868. 3 1.2 4 1.0 2 1.0
869. LOAD COMB 34 GENERATED LOAD_STRENGHT  9
870. 3 0.9 5 1.6
871. LOAD COMB 35 GENERATED LOAD_STRENGHT  10
872. 3 0.9 6 1.6
873. LOAD COMB 36 GENERATED LOAD_STRENGHT  11
874. 3 0.9 7 1.6
875. LOAD COMB 37 GENERATED LOAD_STRENGHT  12
876. 3 0.9 8 1.6
877. LOAD COMB 38 GENERATED LOAD_STRENGHT  13
878. 3 0.9 1 1.0
879. LOAD COMB 39 GENERATED LOAD_STRENGHT  14
880. 3 0.9 2 1.0
881. LOAD COMB 40 GENERATED LOAD_STRENGHT  15
882. 3 1.2 4 1.6 9 1.2
883. LOAD COMB 41 GENERATED LOAD_STRENGHT  16
884. 3 1.2 4 1.6 10 1.2
885. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**747**	**NUMBER OF MEMBERS**	**6**
NUMBER OF PLATES	**685**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**741**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 722/ 31/ 96 DOF
TOTAL PRIMARY LOAD CASES = 10, TOTAL DEGREES OF FREEDOM = 2259
TOTAL LOAD COMBINATION CASES = 31 SO FAR.
SIZE OF STIFFNESS MATRIX = 217 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 18.2/ 87915.0 MB

```
886. START CONCRETE DESIGN
```

```
887. CODE ACI
888. FC 3500 MEMB 279 280 320 TO 323
889. CLB 0.04 MEMB 279 280 320 TO 323
890. CLS 0.04 ALL
891. CLT 0.04 MEMB 279 280 320 TO 323
892. MINSEC 10 ALL
893. MINMAIN 25 ALL
894. MAXMAIN 25 MEMB 320 TO 323
895. DESIGN COLUMN 279 280 320 TO 323
```

==========

COLUMN NO. 279 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
4 - 32 MM	1.287	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

==========

COLUMN NO. 280 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
4 - 32 MM	1.287	1	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

==========

COLUMN NO. 320 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 750.0 X 750.0 MMS, TIED
AREA OF STEEL REQUIRED = 6390.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

16 - 25 MM 1.396 35 STA 0.900
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

==========

COLUMN NO. 321 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 750.0 X 750.0 MMS, TIED
AREA OF STEEL REQUIRED = 6390.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 25 MM	1.396	37	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

==========

COLUMN NO. 322 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 750.0 X 750.0 MMS, TIED
AREA OF STEEL REQUIRED = 5951.2 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 25 MM	1.396	35	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

==========

COLUMN NO. 323 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 750.0 X 750.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 5625.0 SQ. MM

```
BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 12 - 25 MM              1.047        1      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 400.00 MM


********************END OF COLUMN DESIGN RESULTS********************


896. END CONCRETE DESIGN
897. FINISH


          *********** END OF THE STAAD.Pro RUN ***********

            **** DATE= OCT 21,2020   TIME= 17:14:54 ****

      ************************************************************
      *   For technical assistance on STAAD.Pro, please visit    *
      *   http://selectservices.bentley.com/en-US/               *
      *                                                          *
      *   Details about additional assistance from               *
      *   Bentley and Partners can be found at program menu      *
      *   Help->Technical Support                                *
      *                                                          *
      *      Copyright (c) 1997-2014 Bentley Systems, Inc.
      *                 http://www.bentley.com                   *
      ************************************************************
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    OCT 21, 2020                *
*           Time=     9:10:46                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F1-FDN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 13-OCT-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 271 20 -1.5 1.573; 272 20 -0.23 1.573; 273 21.2 -1.5 1.573
     9. 274 21.2 -0.23 1.573; 275 19.5 -1.5 1.573; 276 19.5 -1.5 2.323
    10. 277 19.5 -1.5 0.823; 278 22.7 -1.5 1.573; 280 22.7 -1.5 2.323
    11. 282 22.7 -1.5 0.823; 284 20 -1.5 2.323; 286 21.2 -1.5 2.323; 288 20 -1.5 0.823
    12. 290 21.2 -1.5 0.823; 291 20 -1.5 1.0105; 292 19.5 -1.5 1.0105
    13. 293 20 -1.5 1.198; 294 19.5 -1.5 1.198; 295 20 -1.5 1.3855
    14. 296 19.5 -1.5 1.3855; 297 19.6667 -1.5 0.823; 298 19.6667 -1.5 1.0105
    15. 299 19.8333 -1.5 0.823; 300 19.8333 -1.5 1.0105; 301 19.6667 -1.5 1.198
    16. 302 19.8333 -1.5 1.198; 303 19.6667 -1.5 1.3855; 304 19.8333 -1.5 1.3855
    17. 305 19.6667 -1.5 1.573; 306 19.8333 -1.5 1.573; 307 19.6667 -1.5 1.7605
    18. 308 19.5 -1.5 1.7605; 309 19.8333 -1.5 1.7605; 310 20 -1.5 1.7605
    19. 311 19.6667 -1.5 1.948; 312 19.5 -1.5 1.948; 313 19.8333 -1.5 1.948
    20. 314 20 -1.5 1.948; 315 19.6667 -1.5 2.1355; 316 19.5 -1.5 2.1355
    21. 317 19.8333 -1.5 2.1355; 318 20 -1.5 2.1355; 319 19.6667 -1.5 2.323
    22. 320 19.8333 -1.5 2.323; 321 20.2 -1.5 0.823; 322 20.2 -1.5 1.0105
    23. 323 20.4 -1.5 0.823; 324 20.4 -1.5 1.0105; 325 20.6 -1.5 0.823
    24. 326 20.6 -1.5 1.0105; 327 20.8 -1.5 0.823; 328 20.8 -1.5 1.0105
    25. 329 21 -1.5 0.823; 330 21 -1.5 1.0105; 331 21.2 -1.5 1.0105
    26. 332 20.2 -1.5 1.198; 333 20.4 -1.5 1.198; 334 20.6 -1.5 1.198
    27. 335 20.8 -1.5 1.198; 336 21 -1.5 1.198; 337 21.2 -1.5 1.198
    28. 338 20.2 -1.5 1.3855; 339 20.4 -1.5 1.3855; 340 20.6 -1.5 1.3855
    29. 341 20.8 -1.5 1.3855; 342 21 -1.5 1.3855; 343 21.2 -1.5 1.3855
    30. 344 20.2 -1.5 1.573; 345 20.4 -1.5 1.573; 346 20.6 -1.5 1.573
    31. 347 20.8 -1.5 1.573; 348 21 -1.5 1.573; 349 20.2 -1.5 1.7605
    32. 350 20.4 -1.5 1.7605; 351 20.6 -1.5 1.7605; 352 20.8 -1.5 1.7605
    33. 353 21 -1.5 1.7605; 354 21.2 -1.5 1.7605; 355 20.2 -1.5 1.948
    34. 356 20.4 -1.5 1.948; 357 20.6 -1.5 1.948; 358 20.8 -1.5 1.948
    35. 359 21 -1.5 1.948; 360 21.2 -1.5 1.948; 361 20.2 -1.5 2.1355
    36. 362 20.4 -1.5 2.1355; 363 20.6 -1.5 2.1355; 364 20.8 -1.5 2.1355
    37. 365 21 -1.5 2.1355; 366 21.2 -1.5 2.1355; 367 20.2 -1.5 2.323
    38. 368 20.4 -1.5 2.323; 369 20.6 -1.5 2.323; 370 20.8 -1.5 2.323
```

```
39. 371 21 -1.5 2.323; 372 21.3875 -1.5 0.823; 373 21.3875 -1.5 1.0105
40. 374 21.575 -1.5 0.823; 375 21.575 -1.5 1.0105; 376 21.7625 -1.5 0.823
41. 377 21.7625 -1.5 1.0105; 378 21.95 -1.5 0.823; 379 21.95 -1.5 1.0105
42. 380 22.1375 -1.5 0.823; 381 22.1375 -1.5 1.0105; 382 22.325 -1.5 0.823
43. 383 22.325 -1.5 1.0105; 384 22.5125 -1.5 0.823; 385 22.5125 -1.5 1.0105
44. 386 22.7 -1.5 1.0105; 387 21.3875 -1.5 1.198; 388 21.575 -1.5 1.198
45. 389 21.7625 -1.5 1.198; 390 21.95 -1.5 1.198; 391 22.1375 -1.5 1.198
46. 392 22.325 -1.5 1.198; 393 22.5125 -1.5 1.198; 394 22.7 -1.5 1.198
47. 395 21.3875 -1.5 1.3855; 396 21.575 -1.5 1.3855; 397 21.7625 -1.5 1.3855
48. 398 21.95 -1.5 1.3855; 399 22.1375 -1.5 1.3855; 400 22.325 -1.5 1.3855
49. 401 22.5125 -1.5 1.3855; 402 22.7 -1.5 1.3855; 403 21.3875 -1.5 1.573
50. 404 21.575 -1.5 1.573; 405 21.7625 -1.5 1.573; 406 21.95 -1.5 1.573
51. 407 22.1375 -1.5 1.573; 408 22.325 -1.5 1.573; 409 22.5125 -1.5 1.573
52. 410 21.3875 -1.5 1.7605; 411 21.575 -1.5 1.7605; 412 21.7625 -1.5 1.7605
53. 413 21.95 -1.5 1.7605; 414 22.1375 -1.5 1.7605; 415 22.325 -1.5 1.7605
54. 416 22.5125 -1.5 1.7605; 417 22.7 -1.5 1.7605; 418 21.3875 -1.5 1.948
55. 419 21.575 -1.5 1.948; 420 21.7625 -1.5 1.948; 421 21.95 -1.5 1.948
56. 422 22.1375 -1.5 1.948; 423 22.325 -1.5 1.948; 424 22.5125 -1.5 1.948
57. 425 22.7 -1.5 1.948; 426 21.3875 -1.5 2.1355; 427 21.575 -1.5 2.1355
58. 428 21.7625 -1.5 2.1355; 429 21.95 -1.5 2.1355; 430 22.1375 -1.5 2.1355
59. 431 22.325 -1.5 2.1355; 432 22.5125 -1.5 2.1355; 433 22.7 -1.5 2.1355
60. 434 21.3875 -1.5 2.323; 435 21.575 -1.5 2.323; 436 21.7625 -1.5 2.323
61. 437 21.95 -1.5 2.323; 438 22.1375 -1.5 2.323; 439 22.325 -1.5 2.323
62. 440 22.5125 -1.5 2.323
63. MEMBER INCIDENCES
64. 279 271 272; 280 273 274
65. ELEMENT INCIDENCES SHELL
66. 291 277 297 298 292; 292 297 299 300 298; 293 299 288 291 300
67. 294 292 298 301 294; 295 298 300 302 301; 296 300 291 293 302
68. 297 294 301 303 296; 298 301 302 304 303; 299 302 293 295 304
69. 300 296 303 305 275; 301 303 304 306 305; 302 304 295 271 306
70. 303 275 305 307 308; 304 305 306 309 307; 305 306 271 310 309
71. 306 308 307 311 312; 307 307 309 313 311; 308 309 310 314 313
72. 309 312 311 315 316; 310 311 313 317 315; 311 313 314 318 317
73. 312 316 315 319 276; 313 315 317 320 319; 314 317 318 284 320
74. 315 288 321 322 291; 316 321 323 324 322; 317 323 325 326 324
75. 318 325 327 328 326; 319 327 329 330 328; 320 329 290 331 330
76. 321 291 322 332 293; 322 322 324 333 332; 323 324 326 334 333
77. 324 326 328 335 334; 325 328 330 336 335; 326 330 331 337 336
78. 327 293 332 338 295; 328 332 333 339 338; 329 333 334 340 339
79. 330 334 335 341 340; 331 335 336 342 341; 332 336 337 343 342
80. 333 295 338 344 271; 334 338 339 345 344; 335 339 340 346 345
81. 336 340 341 347 346; 337 341 342 348 347; 338 342 343 273 348
82. 339 271 344 349 310; 340 344 345 350 349; 341 345 346 351 350
83. 342 346 347 352 351; 343 347 348 353 352; 344 348 273 354 353
84. 345 310 349 355 314; 346 349 350 356 355; 347 350 351 357 356
85. 348 351 352 358 357; 349 352 353 359 358; 350 353 354 360 359
86. 351 314 355 361 318; 352 355 356 362 361; 353 356 357 363 362
87. 354 357 358 364 363; 355 358 359 365 364; 356 359 360 366 365
88. 357 318 361 367 284; 358 361 362 368 367; 359 362 363 369 368
89. 360 363 364 370 369; 361 364 365 371 370; 362 365 366 286 371
90. 363 290 372 373 331; 364 372 374 375 373; 365 374 376 377 375
91. 366 376 378 379 377; 367 378 380 381 379; 368 380 382 383 381
92. 369 382 384 385 383; 370 384 282 386 385; 371 331 373 387 337
93. 372 373 375 388 387; 373 375 377 389 388; 374 377 379 390 389
94. 375 379 381 391 390; 376 381 383 392 391; 377 383 385 393 392
```

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 95. 378 385 386 394 393; 379 337 387 395 343; 380 387 388 396 395
 96. 381 388 389 397 396; 382 389 390 398 397; 383 390 391 399 398
 97. 384 391 392 400 399; 385 392 393 401 400; 386 393 394 402 401
 98. 387 343 395 403 273; 388 395 396 404 403; 389 396 397 405 404
 99. 390 397 398 406 405; 391 398 399 407 406; 392 399 400 408 407
100. 393 400 401 409 408; 394 401 402 278 409; 395 273 403 410 354
101. 396 403 404 411 410; 397 404 405 412 411; 398 405 406 413 412
102. 399 406 407 414 413; 400 407 408 415 414; 401 408 409 416 415
103. 402 409 278 417 416; 403 354 410 418 360; 404 410 411 419 418
104. 405 411 412 420 419; 406 412 413 421 420; 407 413 414 422 421
105. 408 414 415 423 422; 409 415 416 424 423; 410 416 417 425 424
106. 411 360 418 426 366; 412 418 419 427 426; 413 419 420 428 427
107. 414 420 421 429 428; 415 421 422 430 429; 416 422 423 431 430
108. 417 423 424 432 431; 418 424 425 433 432; 419 366 426 434 286
109. 420 426 427 435 434; 421 427 428 436 435; 422 428 429 437 436
110. 423 429 430 438 437; 424 430 431 439 438; 425 431 432 440 439
111. 426 432 433 280 440
112. ELEMENT PROPERTY
113. 291 TO 426 THICKNESS 0.45
114. DEFINE MATERIAL START
115. ISOTROPIC CONCRETE
116. E 2.21467E+006
117. POISSON 0.17
118. DENSITY 2.40262
119. ALPHA 1E-005
120. DAMP 0.05
121. TYPE CONCRETE
122. STRENGTH FCU 2812.28
123. END DEFINE MATERIAL
124. MEMBER PROPERTY AMERICAN
125. 279 280 PRIS YD 0.4 ZD 0.4
126. CONSTANTS
127. MATERIAL CONCRETE ALL
128. SUPPORTS
129. 291 TO 426 PLATE MAT DIRECT Y SUBGRADE 4500 PRINT
```

JOINT	INFLUENCE AREA
271	0.034378
273	0.036328
275	0.015628
276	0.007814
277	0.007814
278	0.017578
280	0.008789
282	0.008789
284	0.017189
286	0.018164
288	0.017189
290	0.018164
291	0.034378
292	0.015628
293	0.034378
294	0.015628
295	0.034378

296	0.015628
297	0.015623
298	0.031247
299	0.015623
300	0.031247
301	0.031247
302	0.031247
303	0.031247
304	0.031247
305	0.031247
306	0.031247
307	0.031247
308	0.015628
309	0.031247
310	0.034378
311	0.031247
312	0.015628
313	0.031247
314	0.034378
315	0.031247
316	0.015628
317	0.031247
318	0.034378
319	0.015623
320	0.015623
321	0.018750
322	0.037500
323	0.018750
324	0.037500
325	0.018750
326	0.037500
327	0.018750
328	0.037500
329	0.018750
330	0.037500
331	0.036328
332	0.037500
333	0.037500
334	0.037500
335	0.037500
336	0.037500
337	0.036328
338	0.037500
339	0.037500
340	0.037500
341	0.037500
342	0.037500
343	0.036328
344	0.037500
345	0.037500
346	0.037500
347	0.037500
348	0.037500
349	0.037500
350	0.037500
351	0.037500

352	0.037500
353	0.037500
354	0.036328
355	0.037500
356	0.037500
357	0.037500
358	0.037500
359	0.037500
360	0.036328
361	0.037500
362	0.037500
363	0.037500
364	0.037500
365	0.037500
366	0.036328
367	0.018750
368	0.018750
369	0.018750
370	0.018750
371	0.018750
372	0.017578
373	0.035156
374	0.017578
375	0.035156
376	0.017578
377	0.035156
378	0.017578
379	0.035156
380	0.017578
381	0.035156
382	0.017578
383	0.035156
384	0.017578
385	0.035156
386	0.017578
387	0.035156
388	0.035156
389	0.035156
390	0.035156
391	0.035156
392	0.035156
393	0.035156
394	0.017578
395	0.035156
396	0.035156
397	0.035156
398	0.035156
399	0.035156
400	0.035156
401	0.035156
402	0.017578
403	0.035156
404	0.035156
405	0.035156
406	0.035156
407	0.035156

408	0.035156
409	0.035156
410	0.035156
411	0.035156
412	0.035156
413	0.035156
414	0.035156
415	0.035156
416	0.035156
417	0.017578
418	0.035156
419	0.035156
420	0.035156
421	0.035156
422	0.035156
423	0.035156
424	0.035156
425	0.017578
426	0.035156
427	0.035156
428	0.035156
429	0.035156
430	0.035156
431	0.035156
432	0.035156
433	0.017578
434	0.017578
435	0.017578
436	0.017578
437	0.017578
438	0.017578
439	0.017578
440	0.017578
TOTAL	4.800001

```
130. ***************
131. LOAD 1 LOADTYPE SEISMIC  TITLE SLST(+X)
132. JOINT LOAD
133. 272 FX 0.09
134. 274 FX 0.03
135. 272 FZ 0.06
136. 274 FZ 0.04
137. 272 FY 0.23
138. 274 FY -0.22
139. LOAD 2 LOADTYPE SEISMIC  TITLE SLST(-X)
140. JOINT LOAD
141. 272 FX -0.09
142. 274 FX -0.03
143. 272 FZ -0.06
144. 274 FZ -0.03
145. 272 FY -0.23
146. 274 FY 0.22
147. LOAD 3 LOADTYPE DEAD  TITLE DEAD LOAD
148. SELFWEIGHT Y -1
149. JOINT LOAD
150. 272 FX 0.02
151. 274 FX -0.02
```

```
152. 272 FZ 0.75
153. 274 FZ 1.68
154. 272 FY -3.85
155. 274 FY -5.11
156. ELEMENT LOAD
157. 291 TO 426 PR GY -1.425
158. LOAD 4 LOADTYPE LIVE  TITLE LIVE LOAD
159. JOINT LOAD
160. 272 FX 0.05
161. 274 FX -0.06
162. 272 FZ 1.3
163. 274 FZ 4.04
164. 272 FY -2.9
165. 274 FY -5.66
166. LOAD 5 LOADTYPE WIND  TITLE WIND (+) X
167. JOINT LOAD
168. 272 FX 4.6
169. 274 FX -1.36
170. 272 FZ 0.15
171. 274 FZ -0.19
172. 272 FY 10.42
173. 274 FY -10.39
174. LOAD 6 LOADTYPE WIND  TITLE WIND (-) X
175. JOINT LOAD
176. 272 FX -4.82
177. 274 FX 1.45
178. 272 FZ -0.17
179. 274 FZ 0.21
180. 272 FY -10.69
181. 274 FY 10.66
182. LOAD 7 LOADTYPE WIND  TITLE WIND (+) Z
183. JOINT LOAD
184. 272 FX -0.01
185. 274 FX 0
186. 272 FZ 2.62
187. 274 FZ 2.1
188. 272 FY -2.78
189. 274 FY -2.78
190. LOAD 8 LOADTYPE WIND  TITLE WIND (-) Z
191. JOINT LOAD
192. 272 FX 0.011
193. 274 FX -0.001
194. 272 FZ -2.622
195. 274 FZ -2.102
196. 272 FY 2.778
197. 274 FY 2.782
198. LOAD 9 LOADTYPE TEMPERATURE  TITLE THERMAL LOAD
199. JOINT LOAD
200. 272 FX -0.26
201. 274 FX 0.07
202. 272 FZ -1.68
203. 274 FZ 0.88
204. 272 FY 1.55
205. 274 FY -1.74
206. LOAD 10 LOADTYPE TEMPERATURE  TITLE THERMAL LOAD
207. JOINT LOAD
```

```
208. 272 FX 0.26
209. 274 FX -0.07
210. 272 FZ 1.68
211. 274 FZ -0.88
212. 272 FY -1.55
213. 274 FY 1.74
214. LOAD COMB 11 GENERATED SERVICIBILITY  1
215. 3 1.0
216. LOAD COMB 12 GENERATED SERVICIBILITY  2
217. 3 1.0 4 1.0 9 1.0
218. LOAD COMB 13 GENERATED SERVICIBILITY  3
219. 3 1.0 4 1.0 10 1.0
220. LOAD COMB 14 GENERATED SERVICIBILITY  4
221. 3 1.0 4 0.75 5 0.75
222. LOAD COMB 15 GENERATED SERVICIBILITY  5
223. 3 1.0 4 0.75 6 0.75
224. LOAD COMB 16 GENERATED SERVICIBILITY  6
225. 3 1.0 4 0.75 7 0.75
226. LOAD COMB 17 GENERATED SERVICIBILITY  7
227. 3 1.0 4 0.75 8 0.75
228. LOAD COMB 18 GENERATED SERVICIBILITY  8
229. 3 1.0 4 0.75 1 0.7
230. LOAD COMB 19 GENERATED SERVICIBILITY  9
231. 3 1.0 4 0.75 2 0.7
232. LOAD COMB 20 GENERATED SERVICIBILITY  10
233. 3 0.6 5 1.0
234. LOAD COMB 21 GENERATED SERVICIBILITY  11
235. 3 0.6 6 1.0
236. LOAD COMB 22 GENERATED SERVICIBILITY  12
237. 3 0.6 7 1.0
238. LOAD COMB 23 GENERATED SERVICIBILITY  13
239. 3 0.6 8 1.0
240. LOAD COMB 24 GENERATED SERVICIBILITY  14
241. 3 0.6 1 0.7
242. LOAD COMB 25 GENERATED SERVICIBILITY  15
243. 3 0.6 2 0.7
244. LOAD COMB 26 GENERATED LOAD_STRENGHT  1
245. 3 1.4
246. LOAD COMB 27 GENERATED LOAD_STRENGHT  2
247. 3 1.4 4 1.7
248. LOAD COMB 28 GENERATED LOAD_STRENGHT  3
249. 3 1.2 4 1.0 5 1.6
250. LOAD COMB 29 GENERATED LOAD_STRENGHT  4
251. 3 1.2 4 1.0 6 1.6
252. LOAD COMB 30 GENERATED LOAD_STRENGHT  5
253. 3 1.2 4 1.0 7 1.6
254. LOAD COMB 31 GENERATED LOAD_STRENGHT  6
255. 3 1.2 4 1.0 8 1.6
256. LOAD COMB 32 GENERATED LOAD_STRENGHT  7
257. 3 1.2 4 1.0 1 1.0
258. LOAD COMB 33 GENERATED LOAD_STRENGHT  8
259. 3 1.2 4 1.0 2 1.0
260. LOAD COMB 34 GENERATED LOAD_STRENGHT  9
261. 3 0.9 5 1.6
262. LOAD COMB 35 GENERATED LOAD_STRENGHT  10
263. 3 0.9 6 1.6
```

```
264. LOAD COMB 36 GENERATED LOAD_STRENGHT  11
265. 3 0.9 7 1.6
266. LOAD COMB 37 GENERATED LOAD_STRENGHT  12
267. 3 0.9 8 1.6
268. LOAD COMB 38 GENERATED LOAD_STRENGHT  13
269. 3 0.9 1 1.0
270. LOAD COMB 39 GENERATED LOAD_STRENGHT  14
271. 3 0.9 2 1.0
272. LOAD COMB 40 GENERATED LOAD_STRENGHT  15
273. 3 1.2 4 1.6 9 1.2
274. LOAD COMB 41 GENERATED LOAD_STRENGHT  16
275. 3 1.2 4 1.6 10 1.2
276. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**164**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**136**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**162**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 155/ 15/ 48 DOF
TOTAL PRIMARY LOAD CASES = 10, TOTAL DEGREES OF FREEDOM = 498
TOTAL LOAD COMBINATION CASES = 31 SO FAR.
SIZE OF STIFFNESS MATRIX = 24 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.0/ 87946.4 MB

```
277. START CONCRETE DESIGN
```

```
278. CODE ACI
279. FC 3500 MEMB 279 280
280. CLB 0.04 MEMB 279 280
281. CLS 0.04 ALL
282. CLT 0.04 MEMB 279 280
283. MINSEC 10 ALL
284. MINMAIN 16 ALL
285. DESIGN COLUMN 279 280
```

==========

COLUMN NO. 279 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
AREA OF STEEL REQUIRED = 2067.2 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.508	28	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==========

COLUMN NO. 280 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
AREA OF STEEL REQUIRED = 2192.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.508	30	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

**********END OF COLUMN DESIGN RESULTS**********

286. END CONCRETE DESIGN
287. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 21,2020 TIME= 9:11: 0 ****


```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                    *
************************************************************
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5           *
*           Version  20.07.10.64                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    OCT 21, 2020                 *
*           Time=     9:54:49                     *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F3-FDN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 13-OCT-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 271 20 -1.5 1.573; 272 20 1.07 1.573; 273 21.85 -1.5 1.573
     9. 274 21.85 1.07 1.573; 275 20 -1.5 2.573; 276 21.85 -1.5 2.573
    10. 277 20 -1.5 0.573; 278 21.85 -1.5 0.573; 279 22.925 -1.5 1.573
    11. 280 22.925 -1.5 2.573; 281 22.925 -1.5 0.573; 282 18.925 -1.5 1.573
    12. 283 18.925 -1.5 2.573; 284 18.925 -1.5 0.573; 285 19.1937 -1.5 0.573
    13. 286 19.1937 -1.5 0.823; 287 18.925 -1.5 0.823; 288 19.4625 -1.5 0.573
    14. 289 19.4625 -1.5 0.823; 290 19.7313 -1.5 0.573; 291 19.7313 -1.5 0.823
    15. 292 20 -1.5 0.823; 293 19.1937 -1.5 1.073; 294 18.925 -1.5 1.073
    16. 295 19.4625 -1.5 1.073; 296 19.7313 -1.5 1.073; 297 20 -1.5 1.073
    17. 298 19.1937 -1.5 1.323; 299 18.925 -1.5 1.323; 300 19.4625 -1.5 1.323
    18. 301 19.7313 -1.5 1.323; 302 20 -1.5 1.323; 303 19.1937 -1.5 1.573
    19. 304 19.4625 -1.5 1.573; 305 19.7313 -1.5 1.573; 306 19.1937 -1.5 1.823
    20. 307 18.925 -1.5 1.823; 308 19.4625 -1.5 1.823; 309 19.7313 -1.5 1.823
    21. 310 20 -1.5 1.823; 311 19.1937 -1.5 2.073; 312 18.925 -1.5 2.073
    22. 313 19.4625 -1.5 2.073; 314 19.7313 -1.5 2.073; 315 20 -1.5 2.073
    23. 316 19.1937 -1.5 2.323; 317 18.925 -1.5 2.323; 318 19.4625 -1.5 2.323
    24. 319 19.7313 -1.5 2.323; 320 20 -1.5 2.323; 321 19.1937 -1.5 2.573
    25. 322 19.4625 -1.5 2.573; 323 19.7313 -1.5 2.573; 324 22.1188 -1.5 0.573
    26. 325 22.1188 -1.5 0.823; 326 21.85 -1.5 0.823; 327 22.3875 -1.5 0.573
    27. 328 22.3875 -1.5 0.823; 329 22.6563 -1.5 0.573; 330 22.6563 -1.5 0.823
    28. 331 22.925 -1.5 0.823; 332 22.1188 -1.5 1.073; 333 21.85 -1.5 1.073
    29. 334 22.3875 -1.5 1.073; 335 22.6563 -1.5 1.073; 336 22.925 -1.5 1.073
    30. 337 22.1188 -1.5 1.323; 338 21.85 -1.5 1.323; 339 22.3875 -1.5 1.323
    31. 340 22.6563 -1.5 1.323; 341 22.925 -1.5 1.323; 342 22.1188 -1.5 1.573
    32. 343 22.3875 -1.5 1.573; 344 22.6563 -1.5 1.573; 345 22.1188 -1.5 1.823
    33. 346 21.85 -1.5 1.823; 347 22.3875 -1.5 1.823; 348 22.6563 -1.5 1.823
    34. 349 22.925 -1.5 1.823; 350 22.1188 -1.5 2.073; 351 21.85 -1.5 2.073
    35. 352 22.3875 -1.5 2.073; 353 22.6563 -1.5 2.073; 354 22.925 -1.5 2.073
    36. 355 22.1188 -1.5 2.323; 356 21.85 -1.5 2.323; 357 22.3875 -1.5 2.323
    37. 358 22.6563 -1.5 2.323; 359 22.925 -1.5 2.323; 360 22.1188 -1.5 2.573
    38. 361 22.3875 -1.5 2.573; 362 22.6563 -1.5 2.573; 363 20.2313 -1.5 0.573
```

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39. 364 20.2313 -1.5 0.823; 365 20.4625 -1.5 0.573; 366 20.4625 -1.5 0.823
40. 367 20.6938 -1.5 0.573; 368 20.6938 -1.5 0.823; 369 20.925 -1.5 0.573
41. 370 20.925 -1.5 0.823; 371 21.1563 -1.5 0.573; 372 21.1563 -1.5 0.823
42. 373 21.3875 -1.5 0.573; 374 21.3875 -1.5 0.823; 375 21.6187 -1.5 0.573
43. 376 21.6187 -1.5 0.823; 377 20.2313 -1.5 1.073; 378 20.4625 -1.5 1.073
44. 379 20.6938 -1.5 1.073; 380 20.925 -1.5 1.073; 381 21.1563 -1.5 1.073
45. 382 21.3875 -1.5 1.073; 383 21.6187 -1.5 1.073; 384 20.2313 -1.5 1.323
46. 385 20.4625 -1.5 1.323; 386 20.6938 -1.5 1.323; 387 20.925 -1.5 1.323
47. 388 21.1563 -1.5 1.323; 389 21.3875 -1.5 1.323; 390 21.6187 -1.5 1.323
48. 391 20.2313 -1.5 1.573; 392 20.4625 -1.5 1.573; 393 20.6938 -1.5 1.573
49. 394 20.925 -1.5 1.573; 395 21.1563 -1.5 1.573; 396 21.3875 -1.5 1.573
50. 397 21.6187 -1.5 1.573; 398 20.2313 -1.5 1.823; 399 20.4625 -1.5 1.823
51. 400 20.6938 -1.5 1.823; 401 20.925 -1.5 1.823; 402 21.1563 -1.5 1.823
52. 403 21.3875 -1.5 1.823; 404 21.6187 -1.5 1.823; 405 20.2313 -1.5 2.073
53. 406 20.4625 -1.5 2.073; 407 20.6938 -1.5 2.073; 408 20.925 -1.5 2.073
54. 409 21.1563 -1.5 2.073; 410 21.3875 -1.5 2.073; 411 21.6187 -1.5 2.073
55. 412 20.2313 -1.5 2.323; 413 20.4625 -1.5 2.323; 414 20.6938 -1.5 2.323
56. 415 20.925 -1.5 2.323; 416 21.1563 -1.5 2.323; 417 21.3875 -1.5 2.323
57. 418 21.6187 -1.5 2.323; 419 20.2313 -1.5 2.573; 420 20.4625 -1.5 2.573
58. 421 20.6938 -1.5 2.573; 422 20.925 -1.5 2.573; 423 21.1563 -1.5 2.573
59. 424 21.3875 -1.5 2.573; 425 21.6187 -1.5 2.573
60. MEMBER INCIDENCES
61. 279 271 272; 280 273 274
62. ELEMENT INCIDENCES SHELL
63. 287 284 285 286 287; 288 285 288 289 286; 289 288 290 291 289
64. 290 290 277 292 291; 291 287 286 293 294; 292 286 289 295 293
65. 293 289 291 296 295; 294 291 292 297 296; 295 294 293 298 299
66. 296 293 295 300 298; 297 295 296 301 300; 298 296 297 302 301
67. 299 299 298 303 282; 300 298 300 304 303; 301 300 301 305 304
68. 302 301 302 271 305; 303 282 303 306 307; 304 303 304 308 306
69. 305 304 305 309 308; 306 305 271 310 309; 307 307 306 311 312
70. 308 306 308 313 311; 309 308 309 314 313; 310 309 310 315 314
71. 311 312 311 316 317; 312 311 313 318 316; 313 313 314 319 318
72. 314 314 315 320 319; 315 317 316 321 283; 316 316 318 322 321
73. 317 318 319 323 322; 318 319 320 275 323; 319 278 324 325 326
74. 320 324 327 328 325; 321 327 329 330 328; 322 329 281 331 330
75. 323 326 325 332 333; 324 325 328 334 332; 325 328 330 335 334
76. 326 330 331 336 335; 327 333 332 337 338; 328 332 334 339 337
77. 329 334 335 340 339; 330 335 336 341 340; 331 338 337 342 273
78. 332 337 339 343 342; 333 339 340 344 343; 334 340 341 279 344
79. 335 273 342 345 346; 336 342 343 347 345; 337 343 344 348 347
80. 338 344 279 349 348; 339 346 345 350 351; 340 345 347 352 350
81. 341 347 348 353 352; 342 348 349 354 353; 343 351 350 355 356
82. 344 350 352 357 355; 345 352 353 358 357; 346 353 354 359 358
83. 347 356 355 360 276; 348 355 357 361 360; 349 357 358 362 361
84. 350 358 359 280 362; 351 277 363 364 292; 352 363 365 366 364
85. 353 365 367 368 366; 354 367 369 370 368; 355 369 371 372 370
86. 356 371 373 374 372; 357 373 375 376 374; 358 375 278 326 376
87. 359 292 364 377 297; 360 364 366 378 377; 361 366 368 379 378
88. 362 368 370 380 379; 363 370 372 381 380; 364 372 374 382 381
89. 365 374 376 383 382; 366 376 326 333 383; 367 297 377 384 302
90. 368 377 378 385 384; 369 378 379 386 385; 370 379 380 387 386
91. 371 380 381 388 387; 372 381 382 389 388; 373 382 383 390 389
92. 374 383 333 338 390; 375 302 384 391 271; 376 384 385 392 391
93. 377 385 386 393 392; 378 386 387 394 393; 379 387 388 395 394
94. 380 388 389 396 395; 381 389 390 397 396; 382 390 338 273 397
```

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 95. 383 271 391 398 310; 384 391 392 399 398; 385 392 393 400 399
 96. 386 393 394 401 400; 387 394 395 402 401; 388 395 396 403 402
 97. 389 396 397 404 403; 390 397 273 346 404; 391 310 398 405 315
 98. 392 398 399 406 405; 393 399 400 407 406; 394 400 401 408 407
 99. 395 401 402 409 408; 396 402 403 410 409; 397 403 404 411 410
100. 398 404 346 351 411; 399 315 405 412 320; 400 405 406 413 412
101. 401 406 407 414 413; 402 407 408 415 414; 403 408 409 416 415
102. 404 409 410 417 416; 405 410 411 418 417; 406 411 351 356 418
103. 407 320 412 419 275; 408 412 413 420 419; 409 413 414 421 420
104. 410 414 415 422 421; 411 415 416 423 422; 412 416 417 424 423
105. 413 417 418 425 424; 414 418 356 276 425
106. ELEMENT PROPERTY
107. 287 TO 414 THICKNESS 0.45
108. DEFINE MATERIAL START
109. ISOTROPIC CONCRETE
110. E 2.21467E+006
111. POISSON 0.17
112. DENSITY 2.40262
113. ALPHA 1E-005
114. DAMP 0.05
115. TYPE CONCRETE
116. STRENGTH FCU 2812.28
117. END DEFINE MATERIAL
118. MEMBER PROPERTY AMERICAN
119. 279 280 PRIS YD 0.5 ZD 0.5
120. CONSTANTS
121. MATERIAL CONCRETE ALL
122. SUPPORTS
123. 287 TO 414 PLATE MAT DIRECT Y SUBGRADE 4500 PRINT
```

JOINT	INFLUENCE AREA
271	0.062500
273	0.062512
275	0.031250
276	0.031256
277	0.031250
278	0.031256
279	0.033587
280	0.016794
281	0.016794
282	0.033587
283	0.016794
284	0.016794
285	0.033594
286	0.067188
287	0.033587
288	0.033600
289	0.067200
290	0.033594
291	0.067188
292	0.062500
293	0.067188
294	0.033587
295	0.067200

296	0.067188
297	0.062500
298	0.067188
299	0.033587
300	0.067200
301	0.067188
302	0.062500
303	0.067188
304	0.067200
305	0.067188
306	0.067188
307	0.033587
308	0.067200
309	0.067188
310	0.062500
311	0.067188
312	0.033587
313	0.067200
314	0.067188
315	0.062500
316	0.067188
317	0.033587
318	0.067200
319	0.067188
320	0.062500
321	0.033594
322	0.033600
323	0.033594
324	0.033594
325	0.067188
326	0.062512
327	0.033594
328	0.067188
329	0.033594
330	0.067188
331	0.033587
332	0.067188
333	0.062512
334	0.067188
335	0.067188
336	0.033587
337	0.067188
338	0.062512
339	0.067188
340	0.067188
341	0.033587
342	0.067188
343	0.067188
344	0.067188
345	0.067188
346	0.062512
347	0.067188
348	0.067188
349	0.033587
350	0.067188
351	0.062512

352	0.067188
353	0.067188
354	0.033587
355	0.067188
356	0.062512
357	0.067188
358	0.067188
359	0.033587
360	0.033594
361	0.033594
362	0.033594
363	0.028906
364	0.057812
365	0.028906
366	0.057813
367	0.028906
368	0.057813
369	0.028906
370	0.057813
371	0.028906
372	0.057813
373	0.028900
374	0.057800
375	0.028906
376	0.057812
377	0.057812
378	0.057813
379	0.057813
380	0.057813
381	0.057813
382	0.057800
383	0.057812
384	0.057812
385	0.057813
386	0.057813
387	0.057813
388	0.057813
389	0.057800
390	0.057812
391	0.057812
392	0.057813
393	0.057813
394	0.057813
395	0.057813
396	0.057800
397	0.057812
398	0.057812
399	0.057813
400	0.057813
401	0.057813
402	0.057813
403	0.057800
404	0.057812
405	0.057812
406	0.057813
407	0.057813

```
       408          0.057813
       409          0.057813
       410          0.057800
       411          0.057812
       412          0.057812
       413          0.057813
       414          0.057813
       415          0.057813
       416          0.057813
       417          0.057800
       418          0.057812
       419          0.028906
       420          0.028906
       421          0.028906
       422          0.028906
       423          0.028906
       424          0.028900
       425          0.028906
    TOTAL           8.000003
124. ***************
125. LOAD 1 LOADTYPE SEISMIC  TITLE SLST(+X)
126. JOINT LOAD
127. 272 FX 0.13
128. 274 FX 0.05
129. 272 FZ 0
130. 274 FZ 0.01
131. 272 FY -1.5
132. 274 FY 2.51
133. LOAD 2 LOADTYPE SEISMIC  TITLE SLST(-X)
134. JOINT LOAD
135. 272 FX -0.02
136. 274 FX -0.15
137. 272 FZ 0
138. 274 FZ -0.15
139. 272 FY -2.52
140. 274 FY 12.42
141. LOAD 3 LOADTYPE DEAD  TITLE DEAD LOAD
142. SELFWEIGHT Y -1
143. JOINT LOAD
144. 272 FX 0
145. 274 FX 0
146. 272 FZ 0
147. 274 FZ 0
148. 272 FY -1.33
149. 274 FY -2.17
150. ELEMENT LOAD
151. 287 TO 414 PR GY -1.425
152. LOAD 4 LOADTYPE LIVE  TITLE LIVE LOAD
153. JOINT LOAD
154. 272 FX 0.22
155. 274 FX -0.22
156. 272 FZ 0
157. 274 FZ 0
158. 272 FY -1.66
159. 274 FY -19.47
160. LOAD 5 LOADTYPE WIND  TITLE WIND (+) X
```

```
161. JOINT LOAD
162. 272 FX -0.75
163. 274 FX 3.75
164. 272 FZ 0
165. 274 FZ -0.01
166. 272 FY 6.57
167. 274 FY -6.57
168. LOAD 6 LOADTYPE WIND  TITLE WIND (-) X
169. JOINT LOAD
170. 272 FX 0.75
171. 274 FX -3.75
172. 272 FZ 0
173. 274 FZ 0.01
174. 272 FY -6.57
175. 274 FY 6.57
176. LOAD 7 LOADTYPE WIND  TITLE WIND (+) Z
177. JOINT LOAD
178. 272 FX 0
179. 274 FX -0.05
180. 272 FZ 0.02
181. 274 FZ 0.04
182. 272 FY 0.8
183. 274 FY 0.45
184. LOAD 8 LOADTYPE WIND  TITLE WIND (-) Z
185. JOINT LOAD
186. 272 FX 0
187. 274 FX 0.04
188. 272 FZ -0.018
189. 274 FZ -0.039
190. 272 FY -0.822
191. 274 FY -0.434
192. LOAD 9 LOADTYPE TEMPERATURE  TITLE THERMAL LOAD
193. JOINT LOAD
194. 272 FX 0
195. 274 FX -3.39
196. 272 FZ -0.14
197. 274 FZ -0.56
198. 272 FY -2.62
199. 274 FY 2.62
200. LOAD 10 LOADTYPE TEMPERATURE  TITLE THERMAL LOAD
201. JOINT LOAD
202. 272 FX 0
203. 274 FX 3.39
204. 272 FZ 0.14
205. 274 FZ 0.56
206. 272 FY 2.62
207. 274 FY -2.62
208. LOAD COMB 11 GENERATED SERVICIBILITY  1
209. 3 1.0
210. LOAD COMB 12 GENERATED SERVICIBILITY  2
211. 3 1.0 4 1.0 9 1.0
212. LOAD COMB 13 GENERATED SERVICIBILITY  3
213. 3 1.0 4 1.0 10 1.0
214. LOAD COMB 14 GENERATED SERVICIBILITY  4
215. 3 1.0 4 0.75 5 0.75
216. LOAD COMB 15 GENERATED SERVICIBILITY  5
```

```
217. 3 1.0 4 0.75 6 0.75
218. LOAD COMB 16 GENERATED SERVICIBILITY  6
219. 3 1.0 4 0.75 7 0.75
220. LOAD COMB 17 GENERATED SERVICIBILITY  7
221. 3 1.0 4 0.75 8 0.75
222. LOAD COMB 18 GENERATED SERVICIBILITY  8
223. 3 1.0 4 0.75 1 0.7
224. LOAD COMB 19 GENERATED SERVICIBILITY  9
225. 3 1.0 4 0.75 2 0.7
226. LOAD COMB 20 GENERATED SERVICIBILITY  10
227. 3 0.6 5 1.0
228. LOAD COMB 21 GENERATED SERVICIBILITY  11
229. 3 0.6 6 1.0
230. LOAD COMB 22 GENERATED SERVICIBILITY  12
231. 3 0.6 7 1.0
232. LOAD COMB 23 GENERATED SERVICIBILITY  13
233. 3 0.6 8 1.0
234. LOAD COMB 24 GENERATED SERVICIBILITY  14
235. 3 0.6 1 0.7
236. LOAD COMB 25 GENERATED SERVICIBILITY  15
237. 3 0.6 2 0.7
238. LOAD COMB 26 GENERATED LOAD_STRENGHT  1
239. 3 1.4
240. LOAD COMB 27 GENERATED LOAD_STRENGHT  2
241. 3 1.4 4 1.7
242. LOAD COMB 28 GENERATED LOAD_STRENGHT  3
243. 3 1.2 4 1.0 5 1.6
244. LOAD COMB 29 GENERATED LOAD_STRENGHT  4
245. 3 1.2 4 1.0 6 1.6
246. LOAD COMB 30 GENERATED LOAD_STRENGHT  5
247. 3 1.2 4 1.0 7 1.6
248. LOAD COMB 31 GENERATED LOAD_STRENGHT  6
249. 3 1.2 4 1.0 8 1.6
250. LOAD COMB 32 GENERATED LOAD_STRENGHT  7
251. 3 1.2 4 1.0 1 1.0
252. LOAD COMB 33 GENERATED LOAD_STRENGHT  8
253. 3 1.2 4 1.0 2 1.0
254. LOAD COMB 34 GENERATED LOAD_STRENGHT  9
255. 3 0.9 5 1.6
256. LOAD COMB 35 GENERATED LOAD_STRENGHT  10
257. 3 0.9 6 1.6
258. LOAD COMB 36 GENERATED LOAD_STRENGHT  11
259. 3 0.9 7 1.6
260. LOAD COMB 37 GENERATED LOAD_STRENGHT  12
261. 3 0.9 8 1.6
262. LOAD COMB 38 GENERATED LOAD_STRENGHT  13
263. 3 0.9 1 1.0
264. LOAD COMB 39 GENERATED LOAD_STRENGHT  14
265. 3 0.9 2 1.0
266. LOAD COMB 40 GENERATED LOAD_STRENGHT  15
267. 3 1.2 4 1.6 9 1.2
268. LOAD COMB 41 GENERATED LOAD_STRENGHT  16
269. 3 1.2 4 1.6 10 1.2
270. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	155	NUMBER OF MEMBERS	2
NUMBER OF PLATES	128	NUMBER OF SOLIDS	0
NUMBER OF SURFACES	0	NUMBER OF SUPPORTS	153

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 149/ 16/ 51 DOF
TOTAL PRIMARY LOAD CASES = 10, TOTAL DEGREES OF FREEDOM = 471
TOTAL LOAD COMBINATION CASES = 31 SO FAR.
SIZE OF STIFFNESS MATRIX = 25 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.0/ 87935.5 MB

271. START CONCRETE DESIGN

```
272. CODE ACI
273. FC 3500 MEMB 279 280
274. CLB 0.04 MEMB 279 280
275. CLS 0.04 ALL
276. CLT 0.04 MEMB 279 280
277. MINSEC 10 ALL
278. MINMAIN 16 ALL
279. DESIGN COLUMN 279 280
```

==========

COLUMN NO. 279 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

==========

COLUMN NO. 280 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	1	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

**********END OF COLUMN DESIGN RESULTS**********

280. END CONCRETE DESIGN
281. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 21,2020 TIME= 9:55: 2 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
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